

Corollary 9. The map Φ is surjective if and only if for each i , $1 \leq i \leq n$, the reduced Gröbner basis G contains a polynomial $x_i - h_i$ where $h_i \in k[y_1, \dots, y_m]$.

Examples

- (1) Let $\Phi : \mathbb{Q}[u, v] \rightarrow \mathbb{Q}[x]$ be defined by $\Phi(u) = x^2 + x$ and $\Phi(v) = x^3$. The reduced Gröbner basis G for the ideal $\mathcal{A} = (u - x^2 - x, v - x^3)$ with respect to the lexicographic monomial ordering $x > u > v$ is

$$\begin{aligned} g_1 &= x^2 + x - u, & g_3 &= vx - x - u^2 + u + 2v, \\ g_2 &= ux + x - u - v, & g_4 &= u^3 - 3uv - v^2 - v. \end{aligned}$$

The kernel of Φ is the ideal generated by $G \cap \mathbb{Q}[u, v] = \{g_4\}$. By Corollary 9, we see that Φ is not surjective. The remainder after general polynomial division of x^4 by $\{g_1, g_2, g_3, g_4\}$ is $x + u^2 - u - 2v \notin \mathbb{Q}[u, v]$, so x^4 is not in the image of Φ . The remainder of $x^5 + x$ is $-u^2 + uv + u + 2v \in \mathbb{Q}[u, v]$ so $x^5 + x = \Phi(-u^2 + uv + u + 2v)$ is in the image of Φ , as a quick check will confirm.

- (2) Let $V = \mathcal{Z}(I) \subset \mathbb{C}^3$ and $W = \mathcal{Z}(J) \subset \mathbb{C}^2$ where $I = (xz + y^2 + z^2, xy - xz + yz - 2z^2)$ and $J = (u^3 - uv^2 + v^3)$ as in Example 2 following Corollary 7. Then the map $\varphi : V \rightarrow W$ defined by $\varphi((a, b, c)) = (c, b)$ is a morphism from V to W . To see this, we must check that $(c, b) \in W$ if $(a, b, c) \in V$. Equivalently, by Theorem 6, we must check that the map

$$\tilde{\varphi} : \mathbb{C}[u, v]/(u^3 - uv^2 + v^3) \longrightarrow \mathbb{C}[x, y, z]/(xz + y^2 + z^2, xy - xz + yz - 2z^2)$$

induced by mapping u to z and v to y is a $\mathbb{C}$ -algebra homomorphism. This in turn is equivalent to verifying that $f = z^3 - zy^2 + y^3$ is an element of the ideal I . In this case f is actually an element in the reduced Gröbner basis for I :

$$xy + y^2 + yz - z^2, \quad xz + y^2 + z^2, \quad y^3 - y^2z + z^3,$$

so certainly $f \in I$. (Note that dividing f by the original two generators for I leaves the nonzero remainder f itself, from which it is much less clear that $f \in I$, so it is important to use a Gröbner basis when working in coordinate rings.)

- (3) In the previous example, let $\mathcal{A} = (u - z, v - y, xz + y^2 + z^2, xy - xz + yz - 2z^2) \subset \mathbb{C}[u, v, x, y, z]$ as in Proposition 8. With respect to the lexicographic monomial ordering $x > y > z > u > v$ the reduced Gröbner basis G for $\mathcal{A}$ is

$$xu + u^2 + v^2, \quad xv - u^2 + uv + v^2, \quad y - v, \quad z - u, \quad u^3 - uv^2 + v^3.$$

By Proposition 8, we see that $\ker \tilde{\varphi}$ is generated by $u^3 - uv^2 + v^3 \equiv 0 \pmod{J}$, so $\tilde{\varphi}$ is injective. Since there is no element of the form $x - h(u, v)$ in G , $\tilde{\varphi}$ is not surjective (in fact x is not in the image).

As a final example, we use the determination of the kernel of k -algebra homomorphisms to compute minimal polynomials of elements in simple algebraic field extensions.

Proposition 10. Suppose α is a root of the irreducible polynomial $p(x) \in k[x]$ and $\beta \in k(\alpha)$, say $\beta = f(\alpha)$ for the polynomial $f \in k[x]$. Let G be the reduced Gröbner basis for the ideal $(p, y - f)$ in $k[x, y]$ for the lexicographic monomial ordering $x > y$. Then the minimal polynomial of β over k is the monic polynomial in $G \cap k[y]$.

Proof: The kernel of the k -algebra homomorphism $k[y] \rightarrow k[x]/(p) \cong k(\alpha)$ defined by mapping y first to f and then to β is the principal ideal generated by the minimal polynomial of β in $k[y]$, and the result follows by Proposition 8.

Example

Take $k = \mathbb{Q}$, and let $\beta = 1 + \sqrt[3]{2} + 3\sqrt[3]{4} \in \mathbb{Q}(\sqrt[3]{2})$. Then the ideal $(x^3 - 2, y - (1 + x + 3x^2))$ in $\mathbb{Q}[x, y]$ has reduced Gröbner basis $\{53x - 3y^2 + 7y + 32, y^3 - 3y^2 - 15y - 93\}$ for the lexicographic monomial ordering $x > y$, so the minimal polynomial for β is $y^3 - 3y^2 - 15y - 93$.

EXERCISES

Let R be a commutative ring with $1 \neq 0$ and let k be a field.

1. Prove the converse to Hilbert's Basis Theorem: if the polynomial ring $R[x]$ is Noetherian, then R is Noetherian.
2. Show that each of the following rings are not Noetherian by exhibiting an explicit infinite increasing chain of ideals:
 - (a) the ring of continuous real valued functions on $[0, 1]$,
 - (b) the ring of all functions from any infinite set X to $\mathbb{Z}/2\mathbb{Z}$.
3. Prove that the field $k(x)$ of rational functions over k in the variable x is not a finitely generated k -algebra. (Recall that $k(x)$ is the field of fractions of the polynomial ring $k[x]$. Note that $k(x)$ is a finitely generated *field extension* over k .)
4. Prove that if R is Noetherian, then so is the ring $R[[x]]$ of formal power series in the variable x with coefficients from R (cf. Exercise 3, Section 7.2). [Mimic the proof of Hilbert's Basis Theorem.]
5. (*Fitting's Lemma*) Suppose M is a Noetherian R -module and $\varphi : M \rightarrow M$ is an R -module endomorphism of M . Prove that $\ker(\varphi^n) \cap \text{image}(\varphi^n) = 0$ for n sufficiently large. Show that if φ is surjective, then φ is an isomorphism. [Observe that $\ker(\varphi) \subseteq \ker(\varphi^2) \subseteq \dots$.]
6. Suppose that $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is an exact sequence of R -modules. Prove that M is a Noetherian R -module if and only if M' and M'' are Noetherian R -modules.
7. Prove that submodules, quotient modules, and finite direct sums of Noetherian R -modules are again Noetherian R -modules.
8. If R is a Noetherian ring, prove that M is a Noetherian R -module if and only if M is a finitely generated R -module. (Thus any submodule of a finitely generated module over a Noetherian ring is also finitely generated.)
9. For k a field show that any subring of the polynomial ring $k[x]$ containing k is Noetherian. Give an example to show such subrings need not be U.F.D.s. [If $k \subset R \subseteq k[x]$ and $y \in R - k$ show that $k[x]$ is a finitely generated $k[y]$ -module; then use the previous two exercises. For the second, consider $k[x^2, x^3]$.]
10. Prove that the subring $k[x, x^2y, x^3y^2, \dots, x^i y^{i-1}, \dots]$ of the polynomial ring $k[x, y]$ is *not* a Noetherian ring, hence not a finitely generated k -algebra. (Thus subrings of Noetherian rings need not be Noetherian and subalgebras of finitely generated k -algebras need not be finitely generated.)
11. Suppose R is a commutative ring in which all the prime ideals are finitely generated. This exercise proves that R is Noetherian.

- (a) Prove that if the collection of ideals of R that are not finitely generated is nonempty, then it contains a maximal element I , and that R/I is a Noetherian ring.
- (b) Prove that there are finitely generated ideals J_1 and J_2 containing I with $J_1 J_2 \subseteq I$ and that $J_1 J_2$ is finitely generated. [Observe that I is not a prime ideal.]
- (c) Prove that $I/J_1 J_2$ is a finitely generated R/I -submodule of $J_1/J_1 J_2$. [Use Exercise 8.]
- (d) Show that (c) implies the contradiction that I would be finitely generated over R and deduce that R is Noetherian.
12. Suppose R is a Noetherian ring and S is a finitely generated R -algebra. If $T \subseteq S$ is an R -algebra such that S is a finitely generated T -module, prove that T is a finitely generated R -algebra. [If $s_1, \dots, s_n$ generate S as an R -algebra, and $s'_1, \dots, s'_m$ generate S as a T -module, show that the elements s_i and $s'_j s'_k$ can be written as finite T -linear combinations of the s'_i . If T_0 is the R -subalgebra generated by the coefficients of these linear combinations, show S (hence T_0) is finitely generated (by the s'_i) as a T_0 -module, and conclude that T is finitely generated as an R -algebra.]
13. Verify properties (1) to (10) of the maps $\mathcal{Z}$ and $\mathcal{I}$.
14. Show that the affine algebraic sets in $\mathbb{A}^1$ over any field k are $\emptyset$, k , and finite subsets of k .
15. If $k = \mathbb{F}_2$ and $V = \{(0, 0), (1, 1)\} \subset \mathbb{A}^2$, show that $\mathcal{I}(V)$ is the product ideal $\mathfrak{m}_1 \mathfrak{m}_2$ where $\mathfrak{m}_1 = (x, y)$ and $\mathfrak{m}_2 = (x - 1, y - 1)$.
16. Suppose that V is a finite algebraic set in $\mathbb{A}^n$. If V has m points, prove that $k[V]$ is isomorphic as a k -algebra to k^m . [Use the Chinese Remainder Theorem.]
17. If k is a finite field show that every subset of $\mathbb{A}^n$ is an affine algebraic set.
18. If $k = \mathbb{F}_q$ is the finite field with q elements show that $\mathcal{I}(\mathbb{A}^1) = (x^q - x) \subset k[x]$.
19. For each nonconstant $f \in k[x]$ describe $\mathcal{Z}(f) \subseteq \mathbb{A}^1$ in terms of the unique factorization of f in $k[x]$, and then use this to describe $\mathcal{I}(\mathcal{Z}(f))$. Deduce that $\mathcal{I}(\mathcal{Z}(f)) = (f)$ if and only if f is the product of distinct linear factors in $k[x]$.
20. If f and g are irreducible polynomials in $k[x, y]$ that are not associates (do not divide each other), show that $\mathcal{Z}((f, g))$ is either $\emptyset$ or a finite set in $\mathbb{A}^2$. [If $(f, g) \neq (1)$, show (f, g) contains a nonzero polynomial in $k[x]$ (and similarly a nonzero polynomial in $k[y]$) by letting $R = k[x]$, $F = k(y)$, and applying Gauss's Lemma to show f and g are relatively prime in $F[y]$.]
21. Identify each 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with entries from k with the point (a, b, c, d) in $\mathbb{A}^4$. Show that the group $SL_2(k)$ of matrices of determinant 1 is an algebraic set in $\mathbb{A}^4$.
22. Prove that $SL_n(k)$ is an affine algebraic set in $\mathbb{A}^{n^2}$. [Generalize the preceding exercise.]
23. Let V be any line in $\mathbb{R}^2$ (the zero set of any nonzero linear polynomial $ax + by - c$). Prove that $\mathbb{R}[V]$ is isomorphic as an $\mathbb{R}$ -algebra to the polynomial ring $\mathbb{R}[x]$, and give the corresponding isomorphism from $\mathbb{A}^1$ to V .
24. Let $V = \mathcal{Z}(xy - z) \subseteq \mathbb{A}^3$. Prove that V is isomorphic to $\mathbb{A}^2$ and provide an explicit isomorphism φ and associated k -algebra isomorphism $\tilde{\varphi}$ from $k[V]$ to $k[\mathbb{A}^2]$, along with their inverses. Is $V = \mathcal{Z}(xy - z^2)$ isomorphic to $\mathbb{A}^2$?
25. Suppose $V \subseteq \mathbb{A}^n$ is an affine algebraic set and $f \in k[V]$. The *graph* of f is the collection of points $\{(a_1, \dots, a_n, f(a_1, \dots, a_n))\}$ in $\mathbb{A}^{n+1}$. Prove that the graph of f is an affine algebraic set isomorphic to V . [The morphism in one direction maps $(a_1, \dots, a_n)$ to $(a_1, \dots, a_n, f(a_1, \dots, a_n))$.]

26. Let $V = \mathcal{Z}(xz - y^2, yz - x^3, z^2 - x^2y) \subseteq \mathbb{A}^3$.
- Prove that the map $\varphi : \mathbb{A}^1 \rightarrow V$ defined by $\varphi(t) = (t^3, t^4, t^5)$ is a surjective morphism. [For the surjectivity, if $(x, y, z) \neq (0, 0, 0)$, let $t = y/x$.]
 - Describe the corresponding k -algebra homomorphism $\tilde{\varphi} : k[V] \rightarrow k[\mathbb{A}^1]$ explicitly.
 - Prove that φ is not an isomorphism.
27. Suppose $\varphi : V \rightarrow W$ is a morphism of affine algebraic sets. If W' is an affine algebraic subset of W prove that the preimage $V' = \varphi^{-1}(W')$ of W' in V is an affine algebraic subset of V . If $W' = \mathcal{Z}(I)$ show that $V' = \mathcal{Z}(\tilde{\varphi}(I))$ for the corresponding morphism $\tilde{\varphi} : k[W] \rightarrow k[V]$.
28. Prove that if V and W are affine algebraic sets, then so is $V \times W$ and $k[V \times W] \cong k[V] \otimes_k k[W]$.

The following seven exercises introduce the notion of the *associated primes* of an R -module M . Cf. also Exercises 30–40 in Section 4 and Exercises 25–30 in Section 5.

Definition. A prime ideal P of R is said to be *associated* to the R -module M (sometimes called an *assassin* for M) if P is the annihilator of some element m of M , i.e., if M contains a submodule Rm isomorphic to R/P . The collection of associated primes for M is denoted $\text{Ass}_R(M)$.

When $M = I$ is an ideal in R , it is customary to abuse the terminology and refer instead to the elements of $\text{Ass}_R(R/I)$ (rather than the less interesting collection $\text{Ass}_R(I)$) as the *primes associated to I* . (Cf. Exercises 28–29 in Section 5.)

29. If $R = \mathbb{Z}$ and $M = \mathbb{Z}/n\mathbb{Z}$, show that $\text{Ass}_R(M)$ consists of the prime ideals (p) for the prime divisors p of n .
30. If M is the union of some collection of submodules M_i , prove that $\text{Ass}_R(M)$ is the union of the collection $\text{Ass}_R(M_i)$.
31. Suppose that $\text{Ann}(m) = P$, i.e., that $Rm \cong R/P$. Prove that if $0 \neq m' \in Rm$ then $\text{Ann}(m') = P$. Deduce that $\text{Ass}_R(R/P) = \{P\}$. [Observe that R/P is an integral domain.]
32. Suppose that M is an R -module and that P is a maximal element in the collection of ideals of the form $\text{Ann}(m)$, for $m \in M$. Prove that P is a prime ideal. [If $P = \text{Ann}(m)$ and $ab \in P$, show that $bm \neq 0$ implies $\text{Ann}(m) \subseteq \text{Ann}(bm)$ and use the maximality of P to deduce that $a \in \text{Ann}(bm) = P$.]
33. Suppose R is a Noetherian ring and $M \neq 0$ is an R -module. Prove that $\text{Ass}_R(M) \neq \emptyset$. [Use Exercise 32.]
34. If L is a submodule of M with quotient $N \cong M/L$, prove that there are containments $\text{Ass}_R(N) \subseteq \text{Ass}_R(M) \subseteq \text{Ass}_R(L) \cup \text{Ass}_R(N)$, and show that both containments can be proper. [If $Rm \cong R/P$, show that $Rm \cap L = 0$ implies $P \in \text{Ass}_R(N)$ and if $Rm \cap L \neq 0$ then $P \in \text{Ass}_R(L)$ (by Exercise 31). For the second statement, consider $n\mathbb{Z} \subset \mathbb{Z}$.]
35. Suppose M is an R -module and let $\mathcal{S}$ be a subset of the prime ideals in $\text{Ass}_R(M)$. Prove there is a submodule N of M with $\text{Ass}_R(N) = \mathcal{S}$ and $\text{Ass}_R(M/N) = \text{Ass}_R(M) - \mathcal{S}$. [Consider the collection of submodules N' of M with $\text{Ass}_R(N') \subseteq \mathcal{S}$. Use Exercise 30 and Zorn's Lemma to show that there is a maximal submodule N subject to $\text{Ass}_R(N) \subseteq \mathcal{S}$. If $P \in \text{Ass}_R(M/N)$, there is a submodule $M'/N \cong R/P$. Use the previous exercise to show that $\text{Ass}_R(M') \subseteq \text{Ass}_R(N) \cup \{P\}$ and then use maximality of N to show $P \in \text{Ass}_R(M) - \mathcal{S}$, so that $\text{Ass}_R(M/N) \subseteq \text{Ass}_R(M) - \mathcal{S}$ and $\text{Ass}_R(N) \subseteq \mathcal{S}$. Use the previous exercise again to conclude that equality holds in each.]

Suppose M is a finitely generated module over the commutative ring R with generators $m_1, \dots, m_n$. The *Fitting ideal* $\mathcal{F}_R(M)$ (of level 0) of M (also called a *determinant ideal*) is the ideal in R generated by the determinants of all $n \times n$ matrices $A = (r_{ij})$ where $r_{ij} \in R$ and $r_{i1}m_1 + \dots + r_{in}m_n = 0$ in M , i.e., the rows of A consist of the coefficients in R of relations among the generators m_i (A is called an $n \times n$ “relations matrix” for M). The following five exercises outline some of the properties of the Fitting ideal.

36. (a) Show that the Fitting ideal of M is also the ideal in R generated by all the $n \times n$ minors of all $p \times n$ matrices $A = (r_{ij})$ for $p \geq 1$ whose rows consist of the coefficients in R of relations among the generators m_i .
 (b) Let A be a fixed $p \times n$ matrix as in (a) and let A' be a $p \times n$ matrix obtained from A by any elementary row or column operation. Show that the ideal in R generated by all the $n \times n$ minors of A is the same as the ideal in R generated by all the $n \times n$ minors of A' .
37. Suppose $m_1, \dots, m_n$ and $m'_1, \dots, m'_{n'}$ are two sets of R -module generators for M . Let $\mathcal{F}$ denote the Fitting ideal for M computed using the generators $m_1, \dots, m_n$ and let $\mathcal{F}'$ denote the Fitting ideal for M computed using the generators $m_1, \dots, m_n, m'_1, \dots, m'_{n'}$.
 (a) Show that $m'_s = a_{s'1}m_1 + \dots + a_{s'n}m_n$ for some $a_{s'1}, \dots, a_{s'n} \in R$, and deduce that $(-a_{s'1}, \dots, -a_{s'n}, 0, \dots, 0, 1, 0, \dots, 0)$ is a relation among $m_1, \dots, m_n, m'_1, \dots, m'_{n'}$.
 (b) If $A = (r_{ij})$ is an $n \times n$ matrix whose rows are the coefficients of relations among $m_1, \dots, m_n$ show that $\det A = \det A'$ where A' is an $(n+n') \times (n+n')$ matrix whose rows are the coefficients of relations among $m_1, \dots, m_n, m'_1, \dots, m'_{n'}$. Deduce that $\mathcal{F} \subseteq \mathcal{F}'$. [Use (a) to find a block upper triangular A' having A in the upper left block and the $n' \times n'$ identity matrix in the lower right block.]
 (c) Prove that $\mathcal{F}' \subseteq \mathcal{F}$ and conclude that $\mathcal{F}' = \mathcal{F}$. [Use the previous exercise.]
 (d) Deduce from (c) that the Fitting ideal $\mathcal{F}_R(M)$ of M is an invariant of M that does not depend on the choice of generators for M used to compute it.
38. All modules in this exercise are assumed finitely generated.
 (a) If M can be generated by n elements prove that $\text{Ann}(M)^n \subseteq \mathcal{F}_R(M) \subseteq \text{Ann}(M)$, where $\text{Ann}(M)$ is the annihilator of M in R . [If A is an $n \times n$ relations matrix for M , then $AX = 0$, where X is the column matrix whose entries are $m_1, \dots, m_n$. Multiply by the adjoint of A to deduce that $\det A$ annihilates M .]
 (b) If $M = M_1 \times M_2$ is the direct product of the R -modules M_1 and M_2 prove that $\mathcal{F}_R(M) = \mathcal{F}_R(M_1)\mathcal{F}_R(M_2)$.
 (c) If $M = (R/I_1) \times \dots \times (R/I_n)$ is the direct product of cyclic R -modules for ideals I_i in R prove that $\mathcal{F}_R(M) = I_1 I_2 \dots I_n$.
 (d) If $R = \mathbb{Z}$ and M is a finitely generated abelian group show that $\mathcal{F}_{\mathbb{Z}}(M) = 0$ if M is infinite and $\mathcal{F}_{\mathbb{Z}}(M) = |M|\mathbb{Z}$ if M is finite.
 (e) If I is an ideal in R prove that the image of $\mathcal{F}_R(M)$ in the quotient R/I is $\mathcal{F}_{R/I}(M/IM)$.
 (f) Prove that $\mathcal{F}_R(M/IM) \subseteq (\mathcal{F}_R(M), I) \subseteq R$.
 (g) If $\varphi : M \rightarrow M'$ is a surjective R -module homomorphism prove $\mathcal{F}_R(M) \subseteq \mathcal{F}_R(M')$.
 (h) If $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$ is a short exact sequence of R -modules, prove that $\mathcal{F}_R(L)\mathcal{F}_R(N) \subseteq \mathcal{F}_R(M)$.
 (i) Suppose R is the polynomial ring $k[x, y, z]$ over the field k . Let $M = R/(x, y^2, yz, z^2)$ and let L be the submodule $(x, y, z)/(x, y^2, yz, z^2)$ of M . Prove that $\mathcal{F}_R(M)$ is (x, y^2, yz, z^2) and $\mathcal{F}_R(L)$ is $(x, y, z)^2$. (This shows that in general the Fitting ideal of a submodule L of M need not contain the Fitting ideal for M .)
39. Suppose M is an R -module and that $\varphi : R^n \rightarrow M$ is a surjective R -module homomorphism (i.e., M can be generated by n elements). Let $L = \ker \varphi$. Prove that the image of the

R -module homomorphism from $\bigwedge^n(L) \rightarrow \bigwedge^n(R^n) \cong R$ induced by the inclusion of L in R^n is the Fitting ideal $\mathcal{F}_R(M)$.

40. Suppose R and S are commutative rings, $\varphi : R \rightarrow S$ is a ring homomorphism, M is a finitely generated R -module, and $M' = S \otimes_R M$ is the S -module obtained by extending scalars from R to S . Prove that the Fitting ideal $\mathcal{F}_S(M')$ for M' over S is the extension to S of the Fitting ideal $\mathcal{F}_R(M)$ for M over R .

The following two exercises indicate how the remainder in Theorem 23 of Chapter 9 can be used to effect computations in quotients of polynomial rings.

41. Suppose $\{g_1, \dots, g_m\}$ is a Gröbner basis for the ideal I in $k[x_1, \dots, x_n]$. Prove that the monomials m not divisible by any $LT(g_i)$, $1 \leq i \leq m$, give a k -vector space basis for the quotient $k[x_1, \dots, x_n]/I$.
42. Let $I = (x^3y - xy^2 + 1, x^2y^2 - y^3 - 1)$ as in Example 1 following Proposition 9.26.
- (a) Use the previous exercise to show that $\{1, y, y^2, y^3\}$ is a basis for the k -vector space $k[x, y]/I$.
- (b) Compute the 4×4 multiplication table for the basis vectors in (a).
43. Suppose $K[x_1, \dots, x_n]$ is a polynomial ring in n variables over a field K and k is a subfield of K . If I is an ideal in $k[x_1, \dots, x_n]$, let I' be the ideal generated by I in $K[x_1, \dots, x_n]$.
- (a) If G is a Gröbner basis for the ideal I in $k[x_1, \dots, x_n]$ with respect to some monomial ordering, show that G is also a Gröbner basis for the ideal I' in $K[x_1, \dots, x_n]$ with respect to the same monomial ordering. [Use Buchberger's Criterion.]
- (b) Prove that the dimension of the quotient $k[x_1, \dots, x_n]/I$ as a vector space over k is the same as the dimension of the quotient $K[x_1, \dots, x_n]/I'$ as a vector space over K . [One method: use (a) and Exercise 41.]
- (c) Prove that $I = k[x_1, \dots, x_n]$ if and only if $I' = K[x_1, \dots, x_n]$.
44. Let $V = \mathcal{Z}(x^3 - x^2z - y^2z)$ and $W = \mathcal{Z}(x^2 + y^2 - z^2)$ in $\mathbb{C}^3$. Then $\mathcal{I}(V) = (x^3 - x^2z - y^2z)$ and $\mathcal{I}(W) = (x^2 + y^2 - z^2)$ in $\mathbb{C}[x, y, z]$ (cf. Exercise 23 in Section 3). Show that $\varphi((a, b, c)) = (a^2c - b^2c, 2abc, -a^3)$ defines a morphism from V to W .
45. Let $V = \mathcal{Z}(x^3 + y^3 + 7z^3) \subset \mathbb{C}^3$. Then $\mathcal{I}(V) = (x^3 + y^3 + 7z^3)$ in $\mathbb{C}[x, y, z]$ (cf. Exercise 24 in Section 3).

- (a) Show that

$$\tilde{\varphi}(x) = x(y^3 - 7z^3), \quad \tilde{\varphi}(y) = y(7z^3 - x^3), \quad \tilde{\varphi}(z) = z(x^3 - y^3)$$

defines a $\mathbb{C}$ -algebra homomorphism from $k[V]$ to itself.

- (b) Let $\varphi : V \rightarrow V$ be the morphism corresponding to $\tilde{\varphi}$. Observe that $(-2, 1, 1) \in V$ and compute $\varphi((-2, 1, 1)) \in V$.
- (c) Prove there are infinitely many points (a, b, c) on V with $a, b, c \in \mathbb{Z}$ and the greatest common divisor of a, b , and c is 1.
46. Let $V = \mathcal{Z}(xz + y^2 + z^2, xy - xz + yz - 2z^2) \subset \mathbb{C}^3$ and $W = \mathcal{Z}(u^3 - uv^2 + v^3) \subset \mathbb{C}^2$ as in Example 2 following Corollary 9. Show that the map $\varphi((a, b)) = (-2a^2 + ab, ab - b^2, a^2 - ab)$ defines a morphism from W to V . Show the corresponding $\mathbb{C}$ -algebra homomorphism from $k[V]$ to $k[W]$ has a kernel generated by $x^2 - 3y^2 + yz$.
47. Define $\Phi : \mathbb{Q}[u, v, w] \rightarrow \mathbb{Q}[x, y]$ by $\Phi(u) = x^2 + y$, $\Phi(v) = x + y^2$, and $\Phi(w) = x - y$. Show that neither x nor y is in the image of Φ . Show that $f = 2x^3 - 4xy - 2y^3 - 4y$ is in the image of Φ and find a polynomial in $\mathbb{Q}[u, v, w]$ mapping to f . Show that $\ker \Phi$ is the ideal generated by

$$u^2 - 2uv - 2uw^2 + 4uw + v^2 - 2vw^2 - 4vw + w^4 + 3w^2.$$

48. Suppose α is a root of the irreducible polynomial $p(x) \in k[x]$ and $\beta = f(\alpha)/g(\alpha)$ with polynomials $f(x), g(x) \in k[x]$ where $g(\alpha) \neq 0$.
- (a) Show $ag + bp = 1$ for some polynomials $a, b \in k[x]$ and show $\beta = h(\alpha)$ where $h = af$.
 - (b) Show that the ideals $(p, y - h)$ and $(p, gy - f)$ are equal in $k[x, y]$.
 - (c) Conclude that the minimal polynomial for β is the monic polynomial in $G \cap k[y]$ where G is the reduced Gröbner basis for the ideal $(p, gy - f)$ in $k[x, y]$ for the lexicographic monomial ordering $x > y$.
 - (d) Find the minimal polynomial over $\mathbb{Q}$ of $(3 - \sqrt[3]{2} + \sqrt[3]{4})/(1 + 3\sqrt[3]{2} - 3\sqrt[3]{4})$.

15.2 RADICALS AND AFFINE VARIETIES

Since the zeros of a polynomial f are the same as the zeros of the powers $f^2, f^3, \dots$ in general there are many different ideals in the ring $k[x_1, x_2, \dots, x_n]$ whose zero locus define the same algebraic set V in affine n -space. This leads to the notion of the radical of an ideal, which can be defined in any commutative ring:

Definition. Let I be an ideal in a commutative ring R .

- (1) The *radical* of I , denoted by $\text{rad } I$, is the collection of elements in R some power of which lie in I , i.e.,

$$\text{rad } I = \{a \in R \mid a^k \in I \text{ for some } k \geq 1\}.$$

- (2) The radical of the zero ideal is called the *nilradical* of R .
- (3) An ideal I is called a *radical ideal* if $I = \text{rad } I$.

Note that $a \in R$ is in the nilradical of R if and only if some power of a is 0, so the nilradical of R is the set of all nilpotent elements of R .

Proposition 11. Let I be an ideal in the commutative ring R . Then $\text{rad } I$ is an ideal containing I , and $(\text{rad } I)/I$ is the nilradical of R/I . In particular, R/I has no nilpotent elements if and only if $I = \text{rad } I$ is a radical ideal.

Proof: It is clear that $I \subseteq \text{rad } I$. By definition, the nilradical of R/I consists of the elements in the quotient some power of which is 0. Under the Lattice Isomorphism Theorem for rings this collection of elements corresponds to the elements of R some power of which lie in I , i.e., $\text{rad } I$. It is therefore sufficient to prove that the nilradical N of any commutative ring R is an ideal. Since $0 \in N$, $N \neq \emptyset$. If $a \in N$ and $r \in R$, then since $a^n = 0$ for some $n \geq 1$, the commutativity of R implies that $(ra)^n = r^n a^n = 0$, so $ra \in N$. It remains to see that if $a, b \in N$ then $a + b \in N$. Suppose $a^n = 0$ and $b^m = 0$. Since the Binomial Theorem holds in the commutative ring R (cf. Exercise 25 in Section 7.3),

$$(a + b)^{n+m} = \sum_{i=0}^{n+m} r_i a^i b^{n+m-i}$$

for some ring elements r_i (the binomial coefficients in R). For each term in this sum either $i \geq n$ (in which case $a^i = 0$) or $n + m - i \geq m$, (in which case $b^{n+m-i} = 0$). Hence $(a + b)^{n+m} = 0$, which shows that $a + b$ is nilpotent, i.e., $a + b \in N$.

Proposition 12. The radical of a proper ideal I is the intersection of all prime ideals containing I . In particular, the nilradical is the intersection of all the prime ideals in R .

Proof: Passing to R/I , Proposition 11 shows that it suffices to prove this result for $I = 0$, and in this case the statement is that the nilradical N of R is the intersection of all the prime ideals in R . Let N' denote the intersection of all the prime ideals in R .

Let a be any nilpotent element in R and let P be any prime ideal. Since $a^k = 0$ for some k , there is a smallest positive power n such that $a^n \in P$. Then the product $a^{n-1}a \in P$, and since P is prime, either $a^{n-1} \in P$ or $a \in P$. The former contradicts the minimality of n , and so $a \in P$. Since P was arbitrary, $a \in N'$, which shows that $N \subseteq N'$.

We prove the reverse containment $N' \subseteq N$ by showing that if $a \notin N$, then $a \notin N'$. If a is an element of R not contained in N , let $\mathcal{S}$ be the family of all proper ideals not containing any positive power of a . The collection $\mathcal{S}$ is not empty since $0 \in \mathcal{S}$. Also, if a^k is not contained in any ideal in the chain $I_1 \subseteq I_2 \subseteq \dots$, then a^k is also not contained in the union of these ideals, which shows that chains in $\mathcal{S}$ have upper bounds. By Zorn's Lemma, $\mathcal{S}$ has a maximal element, P . The ideal P must in fact be a prime ideal, as follows. Suppose for some x and y not contained in P , the product xy is an element of P . By the maximality of P , $a^n \in (x) + P$ and $a^m \in (y) + P$ for some positive integers n and m . Then $a^{n+m} \in (xy) + P = P$ contradicting the fact that P is an element of $\mathcal{S}$. This shows that P is indeed a prime ideal not containing a , and hence $a \notin N'$, completing the proof.

Note that in Noetherian rings, Theorem 2 can be used to circumvent the appeal to Zorn's Lemma in the preceding proof.

Corollary 13. Prime (and hence also maximal) ideals are radical.

Proof: If P is a prime ideal, then P is clearly the intersection of all the prime ideals containing P , so $P = \text{rad } P$ by the proposition.

Examples

- (1) In the ring of integers $\mathbb{Z}$, the ideal (a) is a radical ideal if and only if a is square-free or zero. More generally, if $a = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}$ with $a_i \geq 1$ for all i , is the prime factorization of the positive integer a , then $\text{rad}(a) = (p_1 p_2 \cdots p_r)$. For instance, $\text{rad}(180) = (30)$. Note that $(p_1), (p_2), \dots, (p_r)$ are precisely the prime ideals containing the ideal (a) and that their intersection is the ideal $(p_1 p_2 \cdots p_r)$. More generally, in any U.F.D. R , $\text{rad}(a) = (p_1 p_2 \cdots p_r)$ if $a = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}$ is the unique factorization of a into distinct irreducibles.
- (2) The ideal $(x^3 - y^2)$ in $k[x, y]$ is a prime ideal (Exercise 14, Section 9.1), hence is radical.
- (3) If $l_1, \dots, l_m$ are linear polynomials in $k[x_1, x_2, \dots, x_n]$ then $I = (l_1, \dots, l_m)$ is either $k[x_1, x_2, \dots, x_n]$ or a prime ideal, hence I is a radical ideal.

Proposition 14. If R is a Noetherian ring then for any ideal I some positive power of $\text{rad } I$ is contained in I . In particular, the nilradical, N , of a Noetherian ring is a nilpotent ideal: $N^k = 0$ for some $k \geq 1$.

Proof: For any ideal I , the ideal $\text{rad } I$ is finitely generated since R is Noetherian. If $a_1, \dots, a_m$ are generators of $\text{rad } I$, then by definition of the radical, for each i we have $a_i^{k_i} \in I$ for some positive integer k_i . Let k be the maximum of all the k_i . Then the ideal $(\text{rad } I)^{km}$ is generated by elements of the form $a_1^{d_1} a_2^{d_2} \cdots a_m^{d_m}$ where $d_1 + \cdots + d_m = km$, and each of these elements has at least one factor $a_i^{d_i}$ with $d_i \geq k$. Then $a_i^{d_i} \in I$, hence each generator of $(\text{rad } I)^{km}$ lies in I , and so $(\text{rad } I)^{km} \subseteq I$.

The Zariski Topology

We saw in the preceding section that if we restrict to the set of ideals I of $k[\mathbb{A}^n]$ arising as the ideals associated with some algebraic set V , i.e., with $I = \mathcal{I}(V)$, then the maps $\mathcal{Z}$ (from such ideals to algebraic sets) and $\mathcal{I}$ (from algebraic sets to ideals) are inverses of each other: $\mathcal{Z}(\mathcal{I}(V)) = V$ and $\mathcal{I}(\mathcal{Z}(I)) = I$. The elements of the ring $k[\mathbb{A}^n]/\mathcal{I}(V)$ give k -valued functions on V and, since k has no nilpotent elements, powers of nonzero functions are also nonzero functions. Put another way, the ring $k[\mathbb{A}^n]/\mathcal{I}(V)$ has no nilpotent elements, so by Proposition 11, the ideal $\mathcal{I}(V)$ is always a radical ideal.

For arbitrary fields k , it is in general not true that every radical ideal is the ideal of some algebraic set, i.e., of the form $\mathcal{I}(V)$ for some algebraic set V . For example, the ideal $(x^2 + 1)$ in $\mathbb{R}[x]$ is maximal, hence is a radical ideal (by Corollary 13), but is not the ideal of any algebraic set — if it were, then $x^2 + 1$ would have to vanish on that set, but $x^2 + 1$ has no zeros in $\mathbb{R}$. A similar construction works for any field k that is not algebraically closed — there exists an irreducible polynomial $p(x)$ of degree at least 2 in $k[x]$, which then generates the maximal (hence radical) ideal $(p(x))$ in $k[x]$ that has no zeros in k . It is perhaps surprising that the presence of polynomials in one variable that have no zeros is the *only* obstruction to a radical ideal (in *any* number of variables) not being of the form $\mathcal{I}(V)$. This is shown by the next theorem, which provides a fundamental connection between “geometry” and “algebra” and shows that over an *algebraically closed* field (such as $\mathbb{C}$) every radical ideal is of the form $\mathcal{I}(V)$. Over these fields the “geometrically defined” ideals $I = \mathcal{I}(V)$ are therefore the same as the radical ideals, which is a “purely algebraic” property of the ideal I (namely that $I = \text{rad } I$).

Theorem. (*Hilbert’s Nullstellensatz*) Let E be an algebraically closed field. Then $\mathcal{I}(\mathcal{Z}(I)) = \text{rad } I$ for every ideal I of $E[x_1, x_2, \dots, x_n]$. Moreover, the maps $\mathcal{Z}$ and $\mathcal{I}$ in the correspondence

$$\begin{array}{ccc} \{\text{affine algebraic sets}\} & \begin{array}{c} \xrightarrow{\mathcal{I}} \\ \xleftarrow{\mathcal{Z}} \end{array} & \{\text{radical ideals}\} \end{array}$$

are bijections that are inverses of each other.

Proof: This will be proved in the next section (cf. Theorem 32).

Example

The maps $\mathcal{I}$ and $\mathcal{Z}$ in the Nullstellensatz are defined over any field k , and as mentioned are not bijections if k is not algebraically closed. For any field k , however, the map $\mathcal{Z}$ is always surjective and the map $\mathcal{I}$ is always injective (cf. Exercise 9).

One particular consequence of the Nullstellensatz is that for any *proper* ideal I we have $\mathcal{Z}(I) \neq \emptyset$ since $\text{rad } I \neq k[\mathbb{A}^n]$. Hence there always exists at least one common zero (“nullstellen” in German) for all the polynomials contained in a proper ideal (over an algebraically closed field).

We next see that the affine algebraic sets define a topology on affine n -space. Recall that a *topological space* is any set X together with a collection of subsets $\mathcal{T}$ of X , called the *closed sets* in X , satisfying the following axioms:

- (i) an arbitrary intersection of closed sets is closed: if $S_i \in \mathcal{T}$ for i in any index set, then $\bigcap S_i \in \mathcal{T}$,
- (ii) a finite union of closed sets is closed: if $S_1, \dots, S_q \in \mathcal{T}$ then $S_1 \cup \dots \cup S_q \in \mathcal{T}$, and
- (iii) the empty set and the whole space are closed: $\emptyset, X \in \mathcal{T}$.

A subset U of X is called *open* if its complement, $X - U$, is closed (i.e., $X - U \in \mathcal{T}$). The axioms for a topological space are often (equivalently) phrased in terms of the collection of open sets in X .

There are many examples of topological spaces, and a wealth of books on topology. A fixed set X may have a number of different topologies on it, and the collections of closed sets need not be related in these different structures. On any set X there are always at least two topologies: the so-called discrete topology in which every subset of X is closed (i.e., $\mathcal{T}$ is the collection of *all* subsets of X), and the so-called trivial topology in which the only closed sets are $\emptyset$ and X required by axiom (iii).

Suppose now that $X = \mathbb{A}^n$ is affine n -space over an arbitrary field k . Then the collection $\mathcal{T}$ consisting of all the affine algebraic sets in $\mathbb{A}^n$ satisfies the three axioms for a topological space — these are precisely properties (3), (4) and (5) of algebraic sets in the preceding section. It follows that these sets can be taken to be the closed sets in a topology on $\mathbb{A}^n$:

Definition. The *Zariski topology* on affine n -space over an arbitrary field k is the topology in which the closed sets are the affine algebraic sets in $\mathbb{A}^n$.

The Zariski topology is quite “coarse” in the sense that there are “relatively few” closed (or open) sets. For example, for the Zariski topology on $\mathbb{A}^1$ the only closed sets are $\emptyset$, k and the finite sets (cf. Exercise 14 in Section 1), and so the nonempty open sets are the complements of finite sets. If k is an infinite field it follows that in the Zariski topology any two nonempty open sets in $\mathbb{A}^1$ have nonempty intersection. In the language of point-set topology, the Zariski topology is always T_1 (points are closed sets), but for infinite fields the Zariski topology is never T_2 (Hausdorff), i.e., two distinct points never belong to two disjoint open sets (cf. the exercises). For example, when $k = \mathbb{R}$, a nonempty Zariski open set is just the real line $\mathbb{R}$ with some finite number of points removed, and any two such sets have (infinitely many) points in common. Note also that the Zariski open (respectively, closed) sets in $\mathbb{R}$ are also open (respectively, closed) sets with respect to the usual Euclidean topology. The converse is not true; for example the interval $[0, 1]$ is closed in the Euclidean topology but is not closed in the Zariski topology. In this sense the Euclidean topology on $\mathbb{R}$ is much “finer”; there are

many more open sets in the Euclidean topology, in fact the collection of Euclidean open (respectively, closed) sets properly contains the collection of Zariski open (respectively, closed) sets.

The Zariski topology on $\mathbb{A}^n$ is defined so that the affine algebraic subsets of $\mathbb{A}^n$ are closed. In other words, the topology is defined by the zero sets of the ideals in the coordinate ring of $\mathbb{A}^n$. A similar definition can be used to define a Zariski topology on *any* algebraic set V in $\mathbb{A}^n$, as follows. If $k[V]$ is the coordinate ring of V , then the distinct elements of $k[V]$ define distinct k -valued functions on V and there is a natural way of defining

$$\begin{aligned}\mathcal{Z} : \{ \text{ideals in } k[V] \} &\longrightarrow \{ \text{algebraic subsets of } V \} \\ \mathcal{I} : \{ \text{subsets of } V \} &\longrightarrow \{ \text{ideals in } k[V] \}\end{aligned}$$

just as for the case $V = \mathbb{A}^n$. For example, if $\bar{J}$ is an ideal in $k[V]$, then $\mathcal{Z}(\bar{J})$ is the set of elements in V that are common zeros of all the functions in the ideal $\bar{J}$. It is easy to verify that the resulting zero sets in V satisfy the three axioms for a topological space, defining a *Zariski topology* on V , where the closed sets are the algebraic subsets, $\mathcal{Z}(\bar{J})$, for any ideal $\bar{J}$ of $k[V]$. By the Lattice Isomorphism Theorem, the ideals of $k[V]$ are the ideals of $k[x_1, \dots, x_n]$ that contain $\mathcal{I}(V)$ taken mod $\mathcal{I}(V)$. If J is the complete preimage in $k[x_1, \dots, x_n]$ of $\bar{J}$, then the locus of J in $\mathbb{A}^n$ is the same as the locus of $\bar{J}$ in V . It follows that this definition of the Zariski topology on V is just the *subspace topology* for $V \subseteq \mathbb{A}^n$. (Recall that in a topological space X , the closed sets with respect to the subspace topology of a subspace Y are defined to be the sets $C \cap Y$, where C is a closed set in X .) The advantage to the definition of the Zariski topology on V above is that it is defined intrinsically in terms of the coordinate ring $k[V]$ of V , and since the isomorphism type of $k[V]$ does not depend on the affine space $\mathbb{A}^n$ containing V , the Zariski topology on V also depends only on V and not on the ambient affine space in which V may be embedded.

If V and W are two affine algebraic spaces, then since a morphism $\varphi : V \rightarrow W$ is defined by polynomial functions, it is easy to see that φ is *continuous* with respect to the Zariski topologies on V and W (cf. Exercise 27 in Section 1, which shows that the inverse image of a Zariski closed set under a morphism is Zariski closed). In fact the Zariski topology is the coarsest topology in which points are closed and for which polynomial maps are continuous. There exist maps that are continuous with respect to the Zariski topology that are not morphisms, however (cf. Exercise 17).

We have the usual topological notions of closure and density with respect to the Zariski topology.

Definition. For any subset A of $\mathbb{A}^n$, the *Zariski closure* of A is the smallest algebraic set containing A . If $A \subseteq V$ for an algebraic set V then A is *Zariski dense* in V if the Zariski closure of A is V .

For example, if $k = \mathbb{R}$, the algebraic sets in $\mathbb{A}^1$ are $\emptyset$, $\mathbb{R}$, and finite subsets of $\mathbb{R}$ by Exercise 14 in Section 1. The Zariski closure of any infinite set A of real numbers is then all of $\mathbb{A}^1$ and A is Zariski dense in $\mathbb{A}^1$.

Proposition 15. The Zariski closure of a subset A in $\mathbb{A}^n$ is $\mathcal{Z}(\mathcal{I}(A))$.

Proof: Certainly $A \subseteq \mathcal{Z}(\mathcal{I}(A))$. Suppose V is any algebraic set containing A : $A \subseteq V$. Then $\mathcal{I}(V) \subseteq \mathcal{I}(A)$ and $\mathcal{Z}(\mathcal{I}(A)) \subseteq \mathcal{Z}(\mathcal{I}(V)) = V$, so $\mathcal{Z}(\mathcal{I}(A))$ is the smallest algebraic set containing A .

If $\varphi : V \rightarrow W$ is a morphism of algebraic sets, the image $\varphi(V)$ of V need not be an algebraic subset of W , i.e., need not be Zariski closed in W . For example the projection of the hyperbola $V = \mathcal{Z}(xy - 1)$ in $\mathbb{R}^2$ onto the x -axis has image $\mathbb{R}^1 - \{0\}$, which as we have just seen is not an affine algebraic set.

The next result shows that the Zariski closure of the image of a morphism is determined by the kernel of the associated k -algebra homomorphism.

Proposition 16. Suppose $\varphi : V \rightarrow W$ is a morphism of algebraic sets and $\tilde{\varphi} : k[W] \rightarrow k[V]$ is the associated k -algebra homomorphism of coordinate rings. Then

- (1) The kernel of $\tilde{\varphi}$ is $\mathcal{I}(\varphi(V))$.
- (2) The Zariski closure of $\varphi(V)$ is the zero set in W of $\ker \tilde{\varphi}$. In particular, the homomorphism $\tilde{\varphi}$ is injective if and only if $\varphi(V)$ is Zariski dense in W .

Proof: Since $\tilde{\varphi} = f \circ \varphi$, we have $\tilde{\varphi}(f) = 0$ if and only if $(f \circ \varphi)(P) = 0$ for all $P \in V$, i.e., $f(Q) = 0$ for all $Q = \varphi(P) \in \varphi(V)$, which is the statement that $f \in \mathcal{I}(\varphi(V))$, proving the first statement. Since the Zariski closure of $\varphi(V)$ is the zero set of $\mathcal{I}(\varphi(V))$ by the previous proposition, the first statement in (2) follows.

If $\tilde{\varphi}$ is injective then the Zariski closure of $\varphi(V)$ is $\mathcal{Z}(0) = W$ and so $\varphi(V)$ is Zariski dense. Conversely, suppose $\varphi(V)$ is Zariski dense in W , i.e., $\mathcal{Z}(\mathcal{I}(\varphi(V))) = W$. Then $\mathcal{I}(\varphi(V)) = \mathcal{I}(\mathcal{Z}(\mathcal{I}(\varphi(V)))) = \mathcal{I}(W) = 0$ and so $\ker \tilde{\varphi} = 0$.

By Proposition 16 the ideal of polynomials defining the Zariski closure of the image of a morphism φ is the kernel of the corresponding k -algebra homomorphism $\tilde{\varphi}$ in Theorem 6. Proposition 8(1) allows us to compute this kernel using Gröbner bases.

Example: (Implicitization)

A morphism $\varphi : \mathbb{A}^n \rightarrow \mathbb{A}^m$ is just a map

$$\varphi((a_1, a_2, \dots, a_n)) = (\varphi_1(a_1, a_2, \dots, a_n), \dots, \varphi_m(a_1, a_2, \dots, a_n))$$

where φ_i is a polynomial. If k is an infinite field, then $\mathcal{I}(\mathbb{A}^m)$ and $\mathcal{I}(\mathbb{A}^n)$ are both 0, so we may write $k[\mathbb{A}^m] = k[y_1, \dots, y_m]$ and $k[\mathbb{A}^n] = k[x_1, \dots, x_n]$. The k -algebra homomorphism $\tilde{\varphi} : k[\mathbb{A}^m] \rightarrow k[\mathbb{A}^n]$ corresponding to φ is then defined by mapping y_i to $\varphi_i = \varphi_i(x_1, \dots, x_n)$. The image $\varphi(\mathbb{A}^n)$ consists of the set of points $(b_1, \dots, b_m)$ with

$$\begin{aligned} b_1 &= \varphi_1(a_1, a_2, \dots, a_n) \\ b_2 &= \varphi_2(a_1, a_2, \dots, a_n) \\ &\vdots \\ b_m &= \varphi_m(a_1, a_2, \dots, a_n) \end{aligned}$$

where $a_i \in k$. This is the collection of points in $\mathbb{A}^m$ parametrized by the functions $\varphi_1, \dots, \varphi_m$ (with the a_i as parameters). In general such a parametrized collection of points

is not an algebraic set. Finding the equations for the smallest algebraic set containing these points is referred to as *implicitization*, since it amounts to finding a ('smallest') collection of equations satisfied by the b_i (the 'implicit' algebraic relations).

By Proposition 16, this algebraic set is the Zariski closure of $\varphi(\mathbb{A}^n)$ and is the zero set of $\ker \tilde{\varphi}$. By Proposition 8 this kernel is given by $\mathcal{A} \cap k[y_1, \dots, y_m]$, where $\mathcal{A}$ is the ideal in $k[x_1, \dots, x_n, y_1, \dots, y_m]$ generated by the polynomials $y_1 - \varphi_1, \dots, y_m - \varphi_m$. If we compute the reduced Gröbner basis G for $\mathcal{A}$ with respect to the lexicographic monomial ordering $x_1 > \dots > x_n > y_1 > \dots > y_m$, then the polynomials of G lying in $k[y_1, \dots, y_m]$ generate $\ker \tilde{\pi}$. The zero set of these polynomials defines the Zariski closure of $\varphi(\mathbb{A}^n)$ and therefore give the implicitization.

For an explicit example, consider the points $A = \{(a^2, a^3) \mid a \in \mathbb{R}\}$ in $\mathbb{R}^2$. Using coordinates x, y for $\mathbb{R}^2$ and t for $\mathbb{R}^1$, the ideal $\mathcal{A}$ in $\mathbb{R}[x, y, t]$ is $(x - t^2, y - t^3)$. The only element of the reduced Gröbner basis for $\mathcal{A}$ for the ordering $t > x > y$ lying in $\mathbb{R}[x, y]$ is $x^3 - y^2$, so $\mathcal{Z}(x^3 - y^2)$ is the smallest algebraic set in $\mathbb{R}^2$ containing A .

Example: (Projections of Algebraic Sets)

Suppose $V \subseteq \mathbb{A}^n$ is an algebraic set and $m < n$. Let $\pi : V \rightarrow \mathbb{A}^m$ be the morphism projecting onto the first m coordinates:

$$\pi((a_1, a_2, \dots, a_n)) = (a_1, a_2, \dots, a_m).$$

If we use coordinates $x_1, \dots, x_n$ in $k[V]$ and coordinates $y_1, \dots, y_m$ in $k[\mathbb{A}^m]$, the k -algebra homomorphism corresponding to π is given by the map

$$\begin{aligned} \tilde{\pi} : k[y_1, \dots, y_m] &\longrightarrow k[x_1, \dots, x_n]/\mathcal{I}(V) \\ y_i &\longmapsto x_i. \end{aligned}$$

Suppose $V = \mathcal{Z}(I)$ and $I = (f_1, \dots, f_s)$. The Zariski closure of $\pi(V)$ is the zero set of $\ker \tilde{\pi} = \mathcal{A} \cap k[y_1, \dots, y_m]$ where $\mathcal{A}$ is the ideal in $k[x_1, \dots, x_n, y_1, \dots, y_m]$ generated by the polynomials $y_1 - x_1, \dots, y_m - x_m$ together with a set of generators for $\mathcal{I}(V)$. The polynomials involving only $y_1, \dots, y_m$ in the reduced Gröbner basis G for $\mathcal{A}$ with respect to the lexicographic monomial ordering $x_1 > \dots > x_n > y_1 > \dots > y_m$ are generators for the Zariski closure of $\pi(V)$.

If k is algebraically closed we can actually do better with the help of the Nullstellensatz, which gives $\mathcal{I}(V) = \text{rad } I$. Then it is straightforward to see that we obtain the same zero set if in the ideal $\mathcal{A}$ we replace the generators for $\mathcal{I}(V)$ by the generators $f_1, \dots, f_s$ of I (cf. Exercise 46).

For an explicit example, consider projection onto the first two coordinates of $V = \mathcal{Z}(xy - z^2, xz - y, x^2 - z)$ in $\mathbb{C}^3$. Using u, v as coordinates in $\mathbb{C}^2$, we find the reduced Gröbner basis G for the ideal $(u - x, v - y, xy - z^2, xz - y, x^2 - z)$ for the ordering $x > y > z > u > v$ contains only the polynomial $u^3 - v$ in $\mathbb{C}[u, v]$. The smallest algebraic set containing $\pi(V)$ is then the cubic $v = u^3$.

Affine Varieties

We next consider the question of whether an algebraic set can be decomposed into smaller algebraic sets and the corresponding algebraic formulation in terms of its coordinate ring.

Definition. A nonempty affine algebraic set V is called *irreducible* if it cannot be written as $V = V_1 \cup V_2$, where V_1 and V_2 are proper algebraic sets in V . An irreducible affine algebraic set is called an *affine variety*.

Equivalently, an algebraic set (which is a closed set in the Zariski topology) is irreducible if it cannot be written as the union of two proper, closed subsets.

Proposition 17.

- (1) The affine algebraic set V is irreducible if and only if $\mathcal{I}(V)$ is a prime ideal.
- (2) Every nonempty affine algebraic set V may be written uniquely in the form

$$V = V_1 \cup V_2 \cup \cdots \cup V_q$$

where each V_i is irreducible, and $V_i \not\subseteq V_j$ for all $j \neq i$ (i.e., the decomposition is “minimal” or “irredundant”).

Proof: Let $I = \mathcal{I}(V)$ and suppose first that $V = V_1 \cup V_2$ is reducible, where V_1 and V_2 are proper closed subsets. Since $V_1 \neq V$, there is some function f_1 that vanishes on V_1 but not on V , i.e., $f_1 \in \mathcal{I}(V_1) - I$. Similarly, there is a function $f_2 \in \mathcal{I}(V_2) - I$. Then $f_1 f_2$ vanishes on $V_1 \cup V_2 = V$, so $f_1 f_2 \in I$ which shows that I is not a prime ideal. Conversely, if I is not a prime ideal, there exists $f_1, f_2 \in k[\mathbb{A}^n]$ such that $f_1 f_2 \in I$ but neither f_1 nor f_2 belongs to I . Let $V_1 = \mathcal{Z}(f_1) \cap V$ and $V_2 = \mathcal{Z}(f_2) \cap V$. Since the intersection of closed sets is closed, V_1 and V_2 are algebraic sets. Since neither f_1 nor f_2 vanishes on V , both V_1 and V_2 are proper subsets of V . Because $f_1 f_2 \in I$, $V \subseteq \mathcal{Z}(f_1 f_2) = \mathcal{Z}(f_1) \cup \mathcal{Z}(f_2)$, and so V is reducible. This proves (1).

To prove (2), let $\mathcal{S}$ be the collection of nonempty algebraic sets that cannot be written as a finite union of irreducible algebraic sets, and suppose by way of contradiction that $\mathcal{S} \neq \emptyset$. Let I_0 be a maximal element of the corresponding set of ideals, $\{\mathcal{I}(V) \mid V \in \mathcal{S}\}$, which exists (by Theorem 2) since $k[\mathbb{A}^n]$ is Noetherian. Then $V_0 = \mathcal{Z}(I_0)$ is a *minimal* element of $\mathcal{S}$. Since $V_0 \in \mathcal{S}$, it cannot be irreducible by the definition of $\mathcal{S}$. On the other hand, if $V_0 = V_1 \cup V_2$ for some proper, closed subsets V_1, V_2 of V_0 , then by the minimality of V_0 both V_1 and V_2 may be written as finite unions of irreducible algebraic sets. Then V_0 may be written as a finite union of irreducible algebraic sets, a contradiction. This proves $\mathcal{S} = \emptyset$, i.e., every affine algebraic set has a decomposition into affine varieties.

To prove uniqueness, suppose V has two decompositions into affine varieties (where redundant terms have been removed from each decomposition):

$$V = V_1 \cup V_2 \cup \cdots \cup V_r = U_1 \cup U_2 \cup \cdots \cup U_s.$$

Then V_1 is contained in the union of the U_i . Since $V_1 \cap U_i$ is an algebraic set for each i , we obtain a decomposition of V_1 into algebraic subsets:

$$V_1 = (V_1 \cap U_1) \cup (V_1 \cap U_2) \cup \cdots \cup (V_1 \cap U_s).$$

Since V_1 is irreducible, we must have $V_1 = V_1 \cap U_j$ for some j , i.e., $V_1 \subseteq U_j$. By the symmetric argument we have $U_j \subseteq V_{j'}$ for some j' . Thus $V_1 \subseteq V_{j'}$, so $j' = 1$ and $V_1 = U_j$. Applying a similar argument for each V_i it follows that $r = s$ and that $\{V_1, \dots, V_r\} = \{U_1, \dots, U_s\}$. This completes the proof.

Corollary 18. An affine algebraic set V is a variety if and only if its coordinate ring $k[V]$ is an integral domain.

Proof: This follows immediately since $\mathcal{I}(V)$ is a prime ideal if and only if the quotient $k[V] = k[\mathbb{A}^n]/\mathcal{I}(V)$ is an integral domain (Proposition 13 of Chapter 7).

Definition. If V is a variety, then the field of fractions of the integral domain $k[V]$ is called the field of *rational functions* on V and is denoted by $k(V)$. The *dimension* of a variety V , denoted $\dim V$, is defined to be the transcendence degree of $k(V)$ over k .

Examples

- (1) Single points in $\mathbb{A}^n$ are affine varieties since their corresponding ideals in $k[\mathbb{A}^n]$ are maximal ideals. The coordinate ring of a point is isomorphic to k , which is also the field of rational functions. The dimension of a single point is 0. Any finite set is the union of its single point subsets, and this is its unique decomposition into affine subvarieties.
- (2) The x -axis in $\mathbb{R}^2$ is irreducible since it has coordinate ring $\mathbb{R}[x, y]/(y) \cong \mathbb{R}[x]$, which is an integral domain. Similarly, the y -axis and, more generally, lines in $\mathbb{R}^2$ are also irreducible (cf. Exercise 23 in Section 1). Linear sets in $\mathbb{R}^n$ are affine varieties. The field of rational functions on the x -axis is the quotient field $\mathbb{R}(x)$ of $\mathbb{R}[x]$, which is why $\mathbb{R}(x)$ is called a rational function field. The dimension of the x -axis (or, more generally, any line) is 1.
- (3) The union of the x and y axes in $\mathbb{R}^2$, namely $\mathcal{Z}(xy)$, is not a variety: $\mathcal{Z}(xy) = \mathcal{Z}(x) \cup \mathcal{Z}(y)$ is its unique decomposition into subvarieties. The corresponding coordinate ring $\mathbb{R}[x, y]/(xy)$ contains zero divisors.
- (4) The hyperbola $xy = 1$ in $\mathbb{R}^2$ is a variety since we saw in Section 1 that its coordinate ring is the integral domain $\mathbb{R}[x, 1/x]$. Note that the two disjoint branches of the hyperbola (defined by $x > 0$ and $x < 0$) are not subvarieties (cf. also Exercises 12–13).
- (5) If $V = \mathcal{Z}(l_1, l_2, \dots, l_m)$ is the zero set of linear polynomials $l_1, \dots, l_m$ in $k[x_1, \dots, x_m]$ and $V \neq \emptyset$, then V is an affine variety (called a *linear variety*). Note that determining whether $V \neq \emptyset$ is a linear algebra problem.

We end this section with some general ring-theoretic results that were originally motivated by their connection with decomposition questions in geometry.

Primary Decomposition of Ideals in Noetherian Rings

The second statement in Proposition 17 shows that any ideal of the form $\mathcal{I}(V)$ in $k[\mathbb{A}^n]$ may be written uniquely as a finite intersection of prime ideals, and by Hilbert's Nullstellensatz this applies in particular to all radical ideals when k is algebraically closed. In a large class of commutative rings (including all Noetherian rings) every ideal has a *primary decomposition*, which is a similar decomposition but allows ideals that are analogous to “prime powers” (but see the examples below). This decomposition can be considered as a generalization of the factorization of an integer $n \in \mathbb{Z}$ into the product of prime powers. We shall be primarily concerned with the case of Noetherian rings.

Definition. A proper ideal Q in the commutative ring R is called *primary* if whenever $ab \in Q$ and $a \notin Q$, then $b^n \in Q$ for some positive integer n . Equivalently, if $ab \in Q$ and $a \notin Q$, then $b \in \text{rad } Q$.

Some of the basic properties of primary ideals are given in the following proposition.

Proposition 19. Let R be a commutative ring with 1.

- (1) Prime ideals are primary.
- (2) The ideal Q is primary if and only if every zero divisor in R/Q is nilpotent.
- (3) If Q is primary then $\text{rad } Q$ is a prime ideal, and is the unique smallest prime ideal containing Q .
- (4) If Q is an ideal whose radical is a maximal ideal, then Q is a primary ideal.
- (5) Suppose M is a maximal ideal and Q is an ideal with $M^n \subseteq Q \subseteq M$ for some $n \geq 1$. Then Q is a primary ideal with $\text{rad } Q = M$.

Proof: The first two statements are immediate from the definition of a primary ideal. For (3), suppose $ab \in \text{rad } Q$. Then $a^m b^m = (ab)^m \in Q$, and since Q is primary, either $a^m \in Q$, in which case $a \in \text{rad } Q$, or $(b^m)^n \in Q$ for some positive integer n , in which case $b \in \text{rad } Q$. This proves that $\text{rad } Q$ is a prime ideal, and it follows that $\text{rad } Q$ is the smallest prime ideal containing Q (Proposition 12).

To prove (4) we pass to the quotient ring R/Q ; by (2), it suffices to show that every zero divisor in this quotient ring is nilpotent. We are reduced to the situation where $Q = (0)$ and $M = \text{rad } Q = \text{rad}(0)$, which is the nilradical, is a maximal ideal. Since the nilradical is contained in every prime ideal (Proposition 12), it follows that M is the unique prime ideal, so also the unique maximal ideal. If d were a zero divisor, then the ideal (d) would be a proper ideal, hence contained in a maximal ideal. This implies that $d \in M$, hence every zero divisor is indeed nilpotent.

Finally, suppose $M^n \subseteq Q \subseteq M$ for some $n \geq 1$ where M is a maximal ideal. Then $Q \subseteq M$ so $\text{rad } Q \subseteq \text{rad } M = M$. Conversely, $M^n \subseteq Q$ shows that $M \subseteq \text{rad } Q$, so $\text{rad } Q = M$ is a maximal ideal, and Q is primary by (4).

Definition. If Q is a primary ideal, then the prime ideal $P = \text{rad } Q$ is called the *associated prime* to Q , and Q is said to *belong to* P (or to be *P-primary*).

It is easy to check that a finite intersection of P -primary ideals is again a P -primary ideal (cf. the exercises).

Examples

- (1) The primary ideals in $\mathbb{Z}$ are 0 and the ideals (p^m) for p a prime and $m \geq 1$.
- (2) For any field k , the ideal (x) in $k[x, y]$ is primary since it is a prime ideal. For any $n \geq 1$, the ideal $(x, y)^n$ is primary since it is a power of the maximal ideal (x, y) .
- (3) The ideal $Q = (x^2, y)$ in the polynomial ring $k[x, y]$ is primary since we have $(x, y)^2 \subseteq (x^2, y) \subseteq (x, y)$. Similarly, $Q' = (4, x)$ in $\mathbb{Z}[x]$ is a $(2, x)$ -primary ideal.
- (4) Primary ideals need not be powers of prime ideals. For example, the primary ideal Q in the previous example is not the power of a prime ideal, as follows. If $(x^2, y) = P^k$ for some prime ideal P and some $k \geq 1$, then $x^2, y \in P^k \subseteq P$ so $x, y \in P$. Then $P = (x, y)$, and since $y \notin (x, y)^2$, it would follow that $k = 1$ and $Q = (x, y)$. Since $x \notin (x^2, y)$, this is impossible.
- (5) If R is Noetherian, and Q is a primary ideal belonging to the prime ideal P , then

$$P^m \subseteq Q \subseteq P$$

for some $m \geq 1$ by Proposition 14. If P is a maximal ideal, then the last statement in Proposition 19 shows that the converse also holds. This is not necessarily true if P

is a prime ideal that is *not maximal*. For example, consider the ideal $I = (x^2, xy)$ in $k[x, y]$. Then $(x^2) \subset I \subset (x)$, and (x) is a prime ideal, but I is not primary: $xy \in I$ and $x \notin I$, but no positive power of y is an element of I . This example also shows that an ideal whose radical is prime (but not maximal as in (4) of the proposition) is not necessarily primary.

- (6) Powers of prime ideals need not be primary. For example, consider the quotient ring $R = \mathbb{R}[x, y, z]/(xy - z^2)$, the coordinate ring of the cone $z^2 = xy$ in $\mathbb{R}^3$, and let $P = (\bar{x}, \bar{z})$ be the ideal generated by $\bar{x}$ and $\bar{z}$ in R . This is a prime ideal in R since the quotient is $R/(\bar{x}, \bar{z}) \cong \mathbb{R}[x, y, z]/(x, z) \cong \mathbb{R}[y]$ (because $(xy - z^2) \subset (x, z)$). The ideal

$$P^2 = (\bar{x}^2, \bar{x}\bar{z}, \bar{z}^2) = (\bar{x}^2, \bar{x}\bar{z}, \bar{x}\bar{y}) = \bar{x}(\bar{x}, \bar{y}, \bar{z}),$$

however, is not primary: $\bar{x}\bar{y} = \bar{z}^2 \in P^2$, but $\bar{x} \notin P^2$, and no power of $\bar{y}$ is in P^2 . Note that P^2 is another example of an ideal that is not primary whose radical is prime.

- (7) Suppose R is a U.F.D. If π is an irreducible element of R then it is easy to see that the powers (π^n) for $n = 1, 2, \dots$ are (π) -primary ideals. Conversely, suppose Q is a (π) -primary ideal, and let n be the largest integer with $Q \subseteq (\pi^n)$ (such an integer exists since, for example, $\pi^k \in Q$ for some $k \geq 1$, so $n \leq k$). If q is an element of Q not contained in (π^{n+1}) , then $q = r\pi^n$ for some $r \in R$ and $r \notin (\pi)$. Since $r \notin (\pi)$ and Q is (π) -primary, it follows that $\pi^n \in Q$. This shows that $Q = (\pi^n)$.

In the examples above, the ideal (x^2, xy) in $k[x, y]$ is not a primary ideal, but it can be written as the intersection of primary ideals: $(x^2, xy) = (x) \cap (x, y)^2$.

Definition.

- (1) An ideal I in R has a *primary decomposition* if it may be written as a finite intersection of primary ideals:

$$I = \bigcap_{i=1}^m Q_i \quad Q_i \text{ a primary ideal.}$$

- (2) The primary decomposition above is *minimal* and the Q_i are called the *primary components* of I if

- (a) no primary ideal contains the intersection of the remaining primary ideals, i.e., $Q_i \not\supseteq \bigcap_{j \neq i} Q_j$ for all i , and
- (b) the associated prime ideals are all distinct: $\text{rad } Q_i \neq \text{rad } Q_j$ for $i \neq j$.

We now prove that in a Noetherian ring every proper ideal has a minimal primary decomposition. This result is often called the Lasker–Noether Decomposition Theorem, since it was first proved for polynomial rings by the chess master Emanuel Lasker and the proof was later greatly simplified and generalized by Emmy Noether.

Definition. A proper ideal I in the commutative ring R is said to be *irreducible* if I cannot be written nontrivially as the intersection of two other ideals, i.e., if $I = J \cap K$ with ideals J, K implies that $I = J$ or $I = K$.

It is easy to see that a prime ideal is irreducible (see Exercise 11 in Section 7.4). The ideal $(x, y)^2$ in $k[x, y]$ in Example 2 earlier shows that primary ideals need not

be irreducible since it is the intersection of the ideals $(x) + (x, y)^2 = (x, y^2)$ and $(y) + (x, y)^2 = (y, x^2)$. In a Noetherian ring, however, irreducible ideals are necessarily primary:

Proposition 20. Let R be a Noetherian ring. Then

- (1) every irreducible ideal is primary, and
- (2) every proper ideal in R is a finite intersection of irreducible ideals.

Proof: To prove (1) let Q be an irreducible ideal and suppose that $ab \in Q$ and $b \notin Q$. It is easy to check that for any fixed n the set of elements $x \in R$ with $a^n x \in Q$ is an ideal, A_n , in R . Clearly $A_1 \subseteq A_2 \subseteq \dots$ and since R is Noetherian this ascending chain of ideals must stabilize, i.e., $A_n = A_{n+1} = \dots$ for some $n > 0$. Consider the two ideals $I = (a^n) + Q$ and $J = (b) + Q$ of R , each containing Q . If $y \in I \cap J$ then $y = a^n z + q$ for some $z \in R$ and $q \in Q$. Since $ab \in Q$, it follows that $aJ \subseteq Q$, and in particular $ay \in Q$. Then $a^{n+1}z = ay - aq \in Q$, so $z \in A_{n+1} = A_n$. But $z \in A_n$ means that $a^n z \in Q$, so $y \in Q$. It follows that $I \cap J = Q$. Since Q is irreducible and $(b) + Q \neq Q$ (since $b \notin Q$), we must have $a^n \in Q$, which shows that Q is primary.

The proof of (2) is the same as the proof of the second statement in Proposition 17. Let S be the collection of ideals of R that cannot be written as a finite intersection of irreducible ideals. If S is not empty, then since R is Noetherian, there is a maximal element I in S . Then I is not itself irreducible, so $I = J \cap K$ for some ideals J and K distinct from I . Then $I \subset J$ and $I \subset K$ and the maximality of I implies that neither J nor K is in S . But this means that both J and K can be written as finite intersections of irreducible ideals, hence the same would be true for I . This is a contradiction, so $S = \emptyset$, which completes the proof of the proposition.

It is immediate from the previous proposition that in a Noetherian ring every proper ideal has a primary decomposition. If any of the primary ideals in this decomposition contains the intersection of the remaining primary ideals, then we may simply remove this ideal since this will not change the intersection. Hence we may assume the decomposition satisfies (a) in the definition of a minimal decomposition. Since a finite intersection of P -primary ideals is again P -primary (Exercise 31), replacing the primary ideals in the decomposition with the intersections of all those primary ideals belonging to the same prime, we may also assume the decomposition satisfies (b) in the definition of a minimal decomposition. This proves the first statement of the following:

Theorem 21. (Primary Decomposition Theorem) Let R be a Noetherian ring. Then every proper ideal I in R has a minimal primary decomposition. If

$$I = \bigcap_{i=1}^m Q_i = \bigcap_{i=1}^n Q'_i$$

are two minimal primary decompositions for I then the sets of associated primes in the two decompositions are the same:

$$\{\text{rad } Q_1, \text{rad } Q_2, \dots, \text{rad } Q_m\} = \{\text{rad } Q'_1, \text{rad } Q'_2, \dots, \text{rad } Q'_n\}.$$

Moreover, the primary components Q_i belonging to the minimal elements in this set of associated primes are uniquely determined by I .

Proof: The proof of the uniqueness of the set of associated primes is outlined in the exercises, and the proof of the uniqueness of the primary components associated to the minimal primes will be given in Section 4.

Definition. If I is an ideal in the Noetherian ring R then the associated prime ideals in any primary decomposition of I are called the *associated prime ideals of I* . If an associated prime ideal P of I does not contain any other associated prime ideal of I then P is called an *isolated prime ideal*; the remaining associated prime ideals of I are called *embedded prime ideals*.

The prime ideals associated to an ideal I provide a great deal of information about the ideal I (cf. for example Exercises 41 and 43):

Corollary 22. Let I be a proper ideal in the Noetherian ring R .

- (1) A prime ideal P contains the ideal I if and only if P contains one of the associated primes of I , hence if and only if P contains one of the isolated primes of I , i.e., the isolated primes of I are precisely the minimal elements in the set of all prime ideals containing I . In particular, there are only finitely many minimal elements among the prime ideals containing I .
- (2) The radical of I is the intersection of the associated primes of I , hence also the intersection of the isolated primes of I .
- (3) There are prime ideals $P_1, \dots, P_n$ (not necessarily distinct) containing I such that $P_1 P_2 \cdots P_n \subseteq I$.

Proof: The first statement in (1) is an exercise (cf. Exercise 37), and the remainder of (1) follows. Then (2) follows from (1) and Proposition 12, and (3) follows from (2) and Proposition 14.

The last statement in Theorem 21 states that not only the isolated primes, but also the primary components belonging to the isolated primes, are uniquely determined by I . In general the primary decomposition of an ideal I is itself not unique.

Examples

- (1) Let $I = (x^2, xy)$ in $\mathbb{R}[x, y]$. Then

$$(x^2, xy) = (x) \cap (x, y)^2 = (x) \cap (x^2, y)$$

are two minimal primary decompositions for I . The associated primes for I are (x) and $\text{rad}((x, y)^2) = \text{rad}((x^2, y)) = (x, y)$. The prime (x) is the only isolated prime since $(x) \subset (x, y)$, and (x, y) is an embedded prime. A prime ideal P contains I if and only if P contains (x) . The (x) -primary component of I corresponding to this isolated prime is just (x) and occurs in both primary decompositions; the (x, y) -primary component of I corresponding to this embedded prime is not uniquely determined — it is $(x, y)^2$ in the first decomposition and is (x^2, y) in the second. The radical of I is the isolated prime (x) .

This example illustrates the origin of the terminology: in general the irreducible components of the algebraic space $\mathcal{Z}(I)$ defined by I are the zero sets of the isolated primes for I , and the zero sets of the embedded primes are irreducible subspaces of

these components (so are “embedded” in the irreducible components). In this example, $\mathcal{Z}(I)$ is the set of points with $x^2 = xy = 0$, which is just the y -axis in $\mathbb{R}^2$. There is only one irreducible component of this algebraic space (namely the y -axis), which is the locus for the isolated prime (x) . The locus for the embedded prime (x, y) is the origin $(0, 0)$, which is an irreducible subspace embedded in the y -axis.

- (2) Suppose R is a U.F.D. If $a = p_1^{e_1} \cdots p_m^{e_m}$ is the unique factorization into distinct prime powers of the element $a \in R$, then $(a) = (p_1)^{e_1} \cap \cdots \cap (p_m)^{e_m}$ is the minimal primary decomposition of the principal ideal (a) . The associated primes to (a) are $(p_1), \dots, (p_m)$ and are all isolated. The primary decomposition of ideals is a generalization of the factorization of elements into prime powers. See also Exercise 44 for a characterization of U.F.D.s in terms of minimal primary decompositions.

For any Noetherian ring, an ideal I is radical if and only if the primary components of a minimal primary decomposition of I are all *prime* ideals (in which case this primary decomposition is unique), cf. Exercise 43. This generalizes the observation made previously that Proposition 17 together with Hilbert’s Nullstellensatz shows that any radical ideal in $k[\mathbb{A}^n]$ may be written uniquely as a finite intersection of prime ideals when the field k is algebraically closed — this is the algebraic statement that an algebraic set can be decomposed uniquely into the union of irreducible algebraic sets.

EXERCISES

1. Prove (3) of Corollary 22 directly by considering the collection $\mathcal{S}$ of ideals that do not contain a finite product of prime ideals. [If I is a maximal element in $\mathcal{S}$, show that since I is not prime there are ideals J, K properly containing I (hence not in $\mathcal{S}$) with $JK \subseteq I$.]
2. Let I and J be ideals in the ring R . Prove the following statements:
 - (a) If $I^k \subseteq J$ for some $k \geq 1$ then $\text{rad } I \subseteq \text{rad } J$.
 - (b) If $I^k \subseteq J \subseteq I$ for some $k \geq 1$ then $\text{rad } I = \text{rad } J$.
 - (c) $\text{rad}(IJ) = \text{rad}(I \cap J) = \text{rad } I \cap \text{rad } J$.
 - (d) $\text{rad}(\text{rad } I) = \text{rad } I$.
 - (e) $\text{rad } I + \text{rad } J \subseteq \text{rad}(I + J)$ and $\text{rad}(I + J) = \text{rad}(\text{rad } I + \text{rad } J)$.
3. Prove that the intersection of two radical ideals is again a radical ideal.
4. Let $I = \mathfrak{m}_1 \mathfrak{m}_2$ be the product of the ideals $\mathfrak{m}_1 = (x, y)$ and $\mathfrak{m}_2 = (x - 1, y - 1)$ in $\mathbb{F}_2[x, y]$. Prove that I is a radical ideal. Prove that the ideal $(x^3 - y^2)$ is a radical ideal in $\mathbb{F}_2[x, y]$.
5. If $I = (xy, (x - y)z) \subset k[x, y, z]$ prove that $\text{rad } I = (xy, xz, yz)$. For this ideal prove directly that $\mathcal{Z}(I) = \mathcal{Z}(\text{rad } I)$, that $\mathcal{Z}(I)$ is not irreducible, and that $\text{rad } I$ is not prime.
6. Give an example to show that over a field k that is not algebraically closed the containment $I \subseteq \mathcal{I}(\mathcal{Z}(I))$ can be proper even when I is a radical ideal.
7. Suppose R and S are rings and $\varphi: R \rightarrow S$ is a ring homomorphism. If I is an ideal of R show that $\varphi(\text{rad } I) \subseteq \text{rad}(\varphi(I))$. If in addition φ is surjective and I contains the kernel of φ show that $\varphi(\text{rad } I) = \text{rad}(\varphi(I))$.
8. Suppose the prime ideal P contains the ideal I . Prove that P contains the radical of I .
9. Prove that for any field k the map $\mathcal{Z}$ in the Nullstellensatz is always surjective and the map $\mathcal{I}$ in the Nullstellensatz is always injective. [Use property (10) of the maps $\mathcal{Z}$ and $\mathcal{I}$ in Section 1.] Give examples (over a field k that is not algebraically closed) where $\mathcal{Z}$ is not injective and $\mathcal{I}$ is not surjective.

10. Prove that for k a finite field the Zariski topology is the same as the discrete topology: every subset is closed (and open).
11. Let V be a variety in $\mathbb{A}^n$ and let U_1 and U_2 be two subsets of $\mathbb{A}^n$ that are open in the Zariski topology. Prove that if $V \cap U_1 \neq \emptyset$ and $V \cap U_2 \neq \emptyset$ then $V \cap U_1 \cap U_2 \neq \emptyset$. Conclude that *any* nonempty open subset of a variety is *everywhere dense* in the Zariski topology (i.e., its closure is all of V).
12. Use the fact that nonempty open sets of an affine variety are everywhere dense to prove that an affine variety is connected in the Zariski topology. (A topological space is *connected* if it is not the union of two disjoint, proper, open subsets.)
13. Prove that the affine algebraic set V is connected in the Zariski topology if and only if $k[V]$ is not a direct sum of two nonzero ideals. Deduce from this that a variety is connected in the Zariski topology.
14. Prove that if k is an infinite field, then the varieties in $\mathbb{A}^1$ are the empty set, the whole space, and the one point subsets. What are the varieties in $\mathbb{A}^1$ in the case of a finite field k ?
15. Suppose V is a hypersurface in $\mathbb{A}^n$ and $\mathcal{I}(V) = (f)$ for some nonconstant polynomial $f \in k[x_1, x_2, \dots, x_n]$. Prove that V is a variety if and only if f is irreducible.
16. Suppose $V \subseteq \mathbb{A}^n$ is an affine variety and $f \in k[V]$. Prove that the *graph* of f (cf. Exercise 25 in Section 1) is an affine variety.
17. Prove that any permutation of the elements of a field k is a continuous map from $\mathbb{A}^1$ to itself in the Zariski topology on $\mathbb{A}^1$. Deduce that if k is an infinite field, there are Zariski continuous maps from $\mathbb{A}^1$ to itself that are not polynomials.
18. Let V be an affine algebraic set in $\mathbb{A}^n$ over $k = \mathbb{C}$.
 - (a) Prove that morphisms of algebraic sets over $\mathbb{C}$ are continuous in the Euclidean topology (the topology on $\mathbb{C}^n$ obtained by identifying $\mathbb{C}^n$ with $\mathbb{R}^{2n}$ with its usual Euclidean topology).
 - (b) Prove that V is a closed set in the Euclidean topology on $\mathbb{C}^n$ (so the Zariski closed sets of $\mathbb{A}^n$ over $\mathbb{C}$ are also Euclidean closed).
 - (c) Give an example of a set that is closed in the Euclidean topology but is not closed in the Zariski topology, i.e., is not an affine algebraic set (so the Euclidean topology is “finer” than the Zariski topology).
19. Give an example of an injective k -algebra homomorphism $\tilde{\varphi} : k[W] \rightarrow k[V]$ whose associated morphism $\varphi : V \rightarrow W$ is not surjective.
20. Suppose $\varphi : V \rightarrow W$ is a surjective morphism of affine algebraic sets. Prove that if V is a variety then W is a variety.
21. Let V be an algebraic set in $\mathbb{A}^n$ and let $f \in k[V]$. Define $V_f = \{v \in V \mid f(v) \neq 0\}$.
 - (a) Show that V_f is a Zariski open set in V (called a *principal open set* in V).
 - (b) Let J be the ideal in $k[x_1, \dots, x_n, x_{n+1}]$ generated by $\mathcal{I}(V)$ and $x_{n+1}f - 1$, and let $W = \mathcal{Z}(J) \subseteq \mathbb{A}^{n+1}$. Show that $J = \mathcal{I}(W)$ and that the map $\pi : \mathbb{A}^{n+1} \rightarrow \mathbb{A}^n$ by projection onto the first n coordinates is a Zariski continuous bijection from W onto V_f (so the principal open set V_f in V may be embedded as a *closed* set in some (larger) affine space).
 - (c) If U is any open set in V show that $U = V_{f_1} \cup \dots \cup V_{f_m}$ for some $f_1, \dots, f_m \in k[V]$. (This shows that the principal open sets form a *base* for the Zariski topology.)
22. Prove that $GL_n(k)$ is an open affine algebraic set in $\mathbb{A}^{n^2}$ and can be embedded as a closed affine algebraic set in $\mathbb{A}^{n^2+1}$. In particular, deduce that the set $k^\times$ of nonzero elements in

$\mathbb{A}^1$ embeds into $\mathbb{A}^2$ as the hyperbola $xy = 1$. [Use the preceding exercise.]

23. Show that if k is infinite then $\{(a, a^2, a^3) \mid a \in k\} \subset \mathbb{A}^3$ is an affine algebraic variety. If k is finite show that this set is always reducible.
24. Let $V = \mathcal{Z}(xz - y^2, yz - x^3, z^2 - x^2y) \subset \mathbb{A}^3$. Show that if k is infinite then V is an affine variety. [Use Exercise 26 of Section 1 and Exercise 20.]
25. Suppose $f(x) = x^3 + ax^2 + bx + c$ is an irreducible cubic in $\mathbb{Q}[x]$ of discriminant D . Let $I = (x + y + z + a, xy + xz + yz - b, xyz + c)$ in $\mathbb{Q}[x, y, z]$.
 - (a) Prove that I is a prime ideal if and only if D is not a square in $\mathbb{Q}$, in which case I is a maximal ideal and $\mathbb{Q}[x, y, z]/I$ is a splitting field for $f(x)$ over $\mathbb{Q}$.
 - (b) If $D = r^2$, prove that the primary decomposition of I is $I = Q_+ \cap Q_-$ where $Q_{\pm} = (I, (x - y)(x - z)(y - z) \pm r)$. Prove Q_+ and Q_- are maximal ideals, and $\mathbb{Q}[x, y, z]$ modulo Q_+ or Q_- is a splitting field for $f(x)$ over $\mathbb{Q}$.
26. A topological space X is called *quasicompact* if whenever any collection of closed subsets V_i of X has empty intersection, then some finite number of these also has empty intersection, i.e.,

$$\text{whenever } \bigcap_i V_i = \emptyset \text{ there exists } V_{i_1}, V_{i_2}, \dots, V_{i_N} \text{ such that } \bigcap_{i=1}^N V_{i_i} = \emptyset.$$

Prove that every affine algebraic set is quasicompact. [Translate the definition into a property of ideals in $k[x_1, \dots, x_n]$.] (A quasicompact and Hausdorff space is called *compact*.)

27. When k is an infinite field prove that the Zariski topology on k^2 is not the same as taking the Zariski topology on k and then forming the product topology on $k \times k$. [By Exercise 14 of Section 1, in the product topology on $k \times k$ the Zariski closed sets in $k \times k$ are finite unions of sets of the form $\{a\} \times \{b\}$, $\{a\} \times k$ and $k \times \{b\}$, for any $a, b \in k$.]
28. Prove that each of the following rings have infinitely many minimal prime ideals, and that (0) is not the intersection of any finite number of these (so (0) does not have a primary decomposition in these rings):
 - (a) the infinite direct product ring $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \dots$ (which is a Boolean ring, cf. Exercise 23 in Section 7.4).
 - (b) $k[x_1, x_2, \dots]/(x_1x_2, x_3x_4, \dots, x_{2i-1}x_{2i}, \dots)$, where $x_1, x_2, \dots$ are independent variables over the field k .
29. Suppose that A and B are ideals with $AB \subseteq Q$ for a primary ideal Q . Prove that if $A \not\subseteq Q$ then $B \subset \text{rad } Q$.
30. Let Q be a P -primary ideal and suppose A is an ideal not contained in Q . Define $A' = \{r \in R \mid rA \subseteq Q\}$ to be the elements of R that when multiplied by elements of A give elements of Q . Prove that A' is a P -primary ideal.
31. Prove that if Q_1 and Q_2 are primary ideals belonging to the same prime ideal P , then $Q_1 \cap Q_2$ is a primary ideal belonging to P . Conclude that a finite intersection of P -primary ideals is again P -primary.
32. Prove that if Q_1 and Q_2 are primary ideals belonging to the same *maximal* ideal M , then $Q_1 + Q_2$ and $Q_1 Q_2$ are primary ideals belonging to M . Conclude that finite sums and finite products of M -primary ideals are again M -primary.
33. Let $I = (x^2, xy, xz, yz)$ in $k[x, y, z]$. Prove that a primary decomposition of I is $I = (x, y) \cap (x, z) \cap (x, y, z)^2$, determine the isolated and embedded primes of I , and find $\text{rad } I$.
34. Suppose $\varphi : R \rightarrow S$ is a surjective ring homomorphism. Prove that an ideal Q in R containing the kernel of φ is primary if and only if $\varphi(Q)$ is primary in S , and when this is

- the case the prime associated to $\varphi(Q)$ is the image $\varphi(P)$ of the prime P associated to Q .
35. Suppose $\varphi: R \rightarrow S$ is a ring homomorphism.
 - (a) Suppose I is an ideal of R containing $\ker \varphi$ with minimal primary decomposition $I = Q_1 \cap \cdots \cap Q_m$ with $\text{rad } Q_i = P_i$. If φ is a surjective homomorphism prove that $\varphi(I) = \varphi(Q_1) \cap \cdots \cap \varphi(Q_m)$, where $\text{rad } \varphi(Q_i)$ is given by $\varphi(P_i)$, is a minimal primary decomposition of $\varphi(I)$. [Use the previous exercise.]
 - (b) Suppose I is an ideal of S with minimal primary decomposition $I = Q_1 \cap \cdots \cap Q_m$ with $\text{rad } Q_i = P_i$. Prove that $\varphi^{-1}(I) = \varphi^{-1}(Q_1) \cap \cdots \cap \varphi^{-1}(Q_m)$, where $\text{rad } \varphi^{-1}(Q_i)$ is given by $\varphi^{-1}(P_i)$, is a primary decomposition of $\varphi^{-1}(I)$, and is minimal if φ is surjective.
 36. Let $I = (xy, x - yz)$ in $k[x, y, z]$. Prove that $(x, z) \cap (y^2, x - yz)$ is a minimal primary decomposition of I . [Consider the ring homomorphism $\varphi: k[x, y, z] \rightarrow k[y, z]$ given by mapping x to yz , y to y , and z to z and use the previous exercise.]
 37. Prove that a prime ideal P contains the ideal I if and only if P contains one of the associated primes of a minimal primary decomposition of I . [Use Exercise 3 and Exercise 11 in Section 7.4.]
 38. Show that every associated prime ideal for a radical ideal is isolated. [Suppose that $P_2 = \text{rad } Q_2 \subseteq P_1 = \text{rad } Q_1$ in the decomposition of Theorem 21 for the radical ideal I . Show that if $a \in Q_2 \cap \cdots \cap Q_m \subseteq P_2$ then $a^n \in I$ for some $n \geq 1$, conclude that $a \in Q_1$ and derive a contradiction to the minimality of the primary decomposition.]
 39. Fix an element a in the ring R . For any ideal I in the ring R let $I_a = \{r \in R \mid ar \in I\}$.
 - (a) Prove that I_a is an ideal and $I_a = R$ if and only if $a \in I$.
 - (b) Prove that $(I \cap J)_a = I_a \cap J_a$ for ideals I and J .
 - (c) Suppose that Q is a P -primary ideal and that $a \notin Q$. Prove that Q_a is a P -primary ideal and that $Q_a = Q$ if $a \notin P$.
 40. With notation as in the previous exercise, suppose $I = Q_1 \cap \cdots \cap Q_m$ is a minimal primary decomposition of the ideal I and let P_i be the prime ideal associated to Q_i .
 - (a) Prove that $I_a = (Q_1)_a \cap \cdots \cap (Q_m)_a$ and that $\text{rad}(I_a) = \text{rad}((Q_1)_a) \cap \cdots \cap \text{rad}((Q_m)_a)$.
 - (b) Prove that $\text{rad}(I_a)$ is the intersection of the prime ideals P_i for which $a \notin Q_i$. [Use the previous exercise.]
 - (c) Prove that if $\text{rad}(I_a)$ is a prime ideal then $\text{rad}(I_a) = P_j$ for some j . [Use the fact that prime ideals are irreducible.]
 - (d) For each $i = 1, \dots, m$, prove that $\text{rad}(I_a) = P_i$ for some $a \in R$. [Show there exists an $a \in R$ with $a \notin Q_i$ but $a \in Q_j$ for all $j \neq i$.]
 - (e) Show from (c) and (d) that the associated primes for a minimal primary decomposition are precisely the collection of prime ideals among the ideals $\text{rad}(I_a)$ for $a \in R$, and conclude that they are uniquely determined by I independent of the minimal primary decomposition.
 41. Let $P_1, \dots, P_m$ be the associated prime ideals of the ideal (0) in the Noetherian ring R .
 - (a) Show that $P_1 \cap \cdots \cap P_m$ is the collection of nilpotent elements in R . [Apply Corollary 22 to $I = (0)$.]
 - (b) Show that $P_1 \cup \cdots \cup P_m$ is the collection of zero divisors in R . [Let $I = (0)$ in the previous exercise and show that the set of zero divisors is given by the set $\bigcup_{a \in R - \{0\}} (0)_a = \bigcup_{a \in R - \{0\}} \text{rad}((0)_a)$.]
 42. Suppose R is a Noetherian ring. Prove that R is either an integral domain, has nonzero nilpotent elements, or has at least two minimal prime ideals. [Use the previous exercise.]
 43. Prove that the ideal I in the Noetherian ring R is radical if and only if the primary decomposition

nents of a minimal primary decomposition are all prime ideals, and conclude that in this case the minimal primary decomposition is unique. [If $I = Q_1 \cap \cdots \cap Q_m$ is radical with Q_i a P_i -primary component of a minimal decomposition, show that if $a \in P_1 \cap \cdots \cap P_m$ then some power of a is in I , hence $a \in I$ since I is radical. Deduce that $I = P_1 \cap \cdots \cap P_m$ and show that this is also a minimal primary decomposition, i.e., for any i there exists b with $b \notin P_i$, but $b \in P_j$ for $j \neq i$. If $a \in P_i$, show that $ab \in Q_i$, and that $a \in Q_i$. Conclude that $Q_i = P_i$.]

44. Prove that a Noetherian integral domain R is a U.F.D. if and only if for every $a \in R$ the isolated primes associated to the principal ideal (a) are principal ideals. [See Example 2 following Corollary 22. To prove R is a U.F.D., show that an irreducible $a \in R$ is prime and then follow the proof of Theorem 14 in Section 8.3.]
45. Let R be the ring of all real valued functions on the open interval $(-1, 1)$ that have derivatives of all orders (the ring of C^∞ functions). Let

$$F(x) = \begin{cases} e^{-1/x^4} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

(you may assume $F \in R$ and $F^{(n)}(0) = 0$ for all $n \geq 0$). Let (F) be the principal ideal generated by F and let $A = \text{rad}((F))$. Let M be the (maximal) ideal of all functions in R that are zero at $x = 0$ and let $P = \bigcap_{n=1}^{\infty} M^n$.

- (a) Prove that $M = (x)$ is the ideal generated by the function x in R and that $M^n = (x^n)$ consists of the functions whose first $n - 1$ derivatives vanish at the origin.
- (b) Prove that R is not Noetherian (compare Exercise 33 in Section 7.4). [One approach is the following: Let $G(x)$ be the function that is 0 for $x < 0$ and is equal to $F(x)$ for $x \geq 0$. Let I_n be the ideal of functions in R vanishing for all $x \leq 1/n$. Use translates of $G(x)$ to show that $I_1 \subset I_2 \subset I_3 \subset \cdots$ is an infinite ascending chain.]
- (c) Prove that P consists of the functions all of whose derivatives are zero at $x = 0$ (i.e., the functions whose associated Taylor series at $x = 0$ is identically zero), and that P is a prime ideal.
- (d) Prove that $F \in P$ and deduce that $A \subseteq P$.
- (e) Prove that $A \neq P$. [Let $G(x) = e^{-1/x^2}$ when $x \neq 0$ and $G(0) = 0$. Show that $G \in P$ but $G \notin A$.]
- (f) Show that there is a prime ideal Q containing (F) with $Q \neq P, M$. Prove that $Q \subset P$ i.e., there are nonzero prime ideals properly contained in P .
46. Let $\mathcal{A}$ be any ideal in $R = k[x_1, \dots, x_n, y_1, \dots, y_m]$.
- (a) Show that $\text{rad}(\mathcal{A} \cap k[y_1, \dots, y_m]) = \text{rad } \mathcal{A} \cap k[y_1, \dots, y_m]$.
- (b) Suppose $(f_1, \dots, f_s)$ is an ideal in $k[x_1, \dots, x_n]$. Let $F_1, \dots, F_t$ be generators for the radical of $(f_1, \dots, f_s)$, computed in $k[x_1, \dots, x_n]$. Suppose J is an ideal in R and let $\mathcal{A} = J + (f_1, \dots, f_s)$, $\mathcal{B} = J + (F_1, \dots, F_t)$ as ideals in R . Prove that $\text{rad } \mathcal{A} = \text{rad } \mathcal{B}$.
- (c) Conclude from (a) and (b) that $\mathcal{A} = (y_1 - x_1, \dots, y_m - x_m, f_1, \dots, f_s) \cap k[y_1, \dots, y_m]$ and $\mathcal{B} = (y_1 - x_1, \dots, y_m - x_m, F_1, \dots, F_t) \cap k[y_1, \dots, y_m]$ have the same zero sets over an algebraically closed field k . [Use Hilbert's Nullstellensatz.]
47. Determine the Zariski closure in $\mathbb{C}^3$ of the points on the curve $\{(a^2, a^3, a^4) \mid a \in \mathbb{C}\}$.
48. Show that $\mathcal{Z}(x^3 - xyz + z^2)$ is the smallest algebraic set in $\mathbb{R}^3$ containing the points $\{(st, s + t, s^2t) \mid s, t \in \mathbb{R}\}$.
49. Show that $\mathcal{Z}(x^3z^2 - 3xy^2z^2 - y^6 - z^4)$ is the smallest algebraic set in $\mathbb{R}^3$ containing the points $\{(s^2 + t^2, st, s^3) \mid s, t \in \mathbb{R}\}$.

50. Find equations defining the Zariski closure of the set of points $\{(s^4, s^3t, st^3, t^4) \mid s, t \in \mathbb{R}\}$.
51. Show that $V = \mathcal{Z}(x^2 - y^2z)$ (the *Whitney umbrella surface*) is the smallest algebraic set in $\mathbb{R}^3$ containing the points $S = \{(st, s, t^2) \mid s, t \in \mathbb{R}\}$. Show that S is not Zariski closed in V (the missing points explain the name for the surface). Do the same over $\mathbb{C}$, but show that in this case $S = V$ is closed.
52. Let $V = \mathcal{Z}(xz^2 - w^3, xw^2 - y^4, y^4z^2 - w^5) \subset \mathbb{C}^4$. Determine the Zariski closure of the image of V under the projection $\pi((x, y, z, w)) = (x, y, z)$.
53. Let $V = \mathcal{Z}(xy - 1)$ in $\mathbb{A}^2$ and let S be the projection of V onto the x -axis in $\mathbb{A}^1$.
- If $k = \mathbb{R}$, show that $\mathcal{I}(V) = (xy - 1) \subset \mathbb{R}[x, y]$ and that $(u - x, xy - 1) \cap \mathbb{R}[u] = 0$ in $\mathbb{R}[x, y, u]$. Use Propositions 8 and 16 to conclude that the Zariski closure of S is $\mathbb{A}^1$ and show that S is not itself closed.
 - If $k = \mathbb{F}_3$, show that $\mathcal{I}(V) = (xy - 1, x^3 - x, y^3 - y) \subset \mathbb{F}_3[x, y]$ and that $(u - x, xy - 1, x^3 - x, y^3 - y) \cap \mathbb{F}_3[u] = (u^2 - 1)$ in $\mathbb{F}_3[x, y, u]$. Use Propositions 8 and 16 to conclude that S is Zariski closed in $\mathbb{A}^1$.
54. Recall the *ideal quotient* $(I : J) = \{r \in R \mid rJ \in I\}$ of two ideals I, J in a ring R (cf. Exercise 34 ff. in Section 9.6). Clearly $I \subseteq (I : J)$.
- Show that $\mathcal{Z}(I) - \mathcal{Z}(J)$, the set of elements of $\mathcal{Z}(I)$ not lying in $\mathcal{Z}(J)$, is contained in $\mathcal{Z}((I : J))$ and conclude that the Zariski closure of $\mathcal{Z}(I) - \mathcal{Z}(J)$ is contained in $\mathcal{Z}((I : J))$.
 - Show that if k is algebraically closed and I is a radical ideal then $\mathcal{Z}((I : J))$ is precisely the Zariski closure of $\mathcal{Z}(I) - \mathcal{Z}(J)$.
 - Show that if V and W are affine algebraic sets then $(\mathcal{I}(V) : \mathcal{I}(W)) = \mathcal{I}(V - W)$.

15.3 INTEGRAL EXTENSIONS AND HILBERT'S NULLSTELLENSATZ

In this section we consider the important concept of an integral extension of rings, which is a generalization to rings of algebraic extensions of fields. This leads to the definition of the “integers” in finite extensions of $\mathbb{Q}$ (the basic subject of the branch of mathematics called algebraic number theory) and is also related to the existence of tangent lines for algebraic curves.

Definition. Suppose R is a subring of the commutative ring S with $1 = 1_S \in R$.

- (1) An element $s \in S$ is *integral over R* if s is the root of a monic polynomial in $R[x]$.
- (2) The ring S is an *integral extension of R* or just *integral over R* if every $s \in S$ is integral over R .
- (3) The *integral closure of R in S* is the set of elements of S that are integral over R .
- (4) The ring R is said to be *integrally closed in S* if R is equal to its integral closure in S . The integral closure of an integral domain R in its field of fractions is called the *normalization of R* . An integral domain is called *integrally closed* or *normal* if it is integrally closed in its field of fractions.

Before giving some examples of integral extensions we prove some basic properties of integral elements analogous to those of algebraic elements over fields.

Proposition 23. Let R be a subring of the commutative ring S with $1 \in R$ and let $s \in S$. Then the following are equivalent:

- (1) s is integral over R ,
- (2) $R[s]$ is a finitely generated R -module (where $R[s]$ is the ring of all R -linear combinations of powers of s), and
- (3) $s \in T$ for some subring T , $R \subseteq T \subseteq S$, that is a finitely generated R -module.

Proof: Suppose first that (1) holds and let s be a root of the monic polynomial $x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in R[x]$. Then

$$s^n = -(a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \cdots + a_0)$$

and so s^n , and then all higher powers of s , can be expressed as R -linear combinations of $s^{n-1}, \dots, s, 1$. Hence $R[s] = R \cdot 1 + Rs + \cdots + Rs^{n-1}$ is finitely generated as an R -module, which gives (2).

If (2) holds, then (3) holds with $T = R[s]$.

Suppose that (3) holds and let $v_1, v_2, \dots, v_n$ be a finite generating set for T . Then for $i = 1, 2, \dots, n$ the element sv_i is an element of T since T is a ring, and so can be written as R -linear combinations of $v_1, \dots, v_n$:

$$sv_i = \sum_{j=1}^n a_{ij}v_j,$$

i.e.,

$$0 = \sum_{j=1}^n (\delta_{ij}s - a_{ij})v_j \quad i = 1, 2, \dots, n$$

where δ_{ij} is the Kronecker delta. If B is the $n \times n$ matrix whose i, j entry is $\delta_{ij}s - a_{ij}$, and v is the $n \times 1$ column vector whose entries are $v_1, \dots, v_n$, then these equations are simply $Bv = 0$. It follows from Cramer's Rule that $(\det B)v_i = 0$ for all i (cf. Exercise 3, Section 11.4). Since $1 \in T$ is an R -linear combination of $v_1, \dots, v_n$, it follows that $\det B = 0$. But $B = sI - A$, where A is the matrix (a_{ij}) . Thus s is a root of the monic polynomial $\det(xI - A) \in R[x]$ (the characteristic polynomial of A), and so s is a root of a monic polynomial with coefficients in R , which gives (1), completing the proof.

Corollary 24. Let $R \subseteq S$ be as in Proposition 23 and let $s, t \in S$.

- (1) If s and t are integral over R then so are $s \pm t$ and st .
- (2) The integral closure of R in S is a subring of S containing R .
- (3) Integrality is transitive: let S be a subring of T ; if T is integral over S and S is integral over R , then T is integral over R .

Proof: Let s and t be integral over R . By Proposition 23 both $R[s]$ and $R[t]$ are finitely generated R -modules, say

$$R[s] = Rs_1 + Rs_2 + \cdots + Rs_n$$

$$R[t] = Rt_1 + Rt_2 + \cdots + Rt_m.$$

Then

$$R[s, t] = Rs_1t_1 + \cdots + Rs_it_i + \cdots + Rs_nt_n$$

' is a ring containing $s \pm t$ and st that is also a finitely generated R -module. Hence $s \pm t$ and st are also integral over R , which proves (1) and also (2).

To prove (3), let $t \in T$. Since t is integral over S , it is the root of some monic polynomial $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in S[x]$. Since $a_i \in S$ is integral over R , each ring $R[a_i]$ is a finitely generated R -module and so the ring $R_1 = R[a_0, a_1, \dots, a_{n-1}]$ is also a finitely generated R -module. Since the monic polynomial $p(x)$ has its coefficients in R_1 , t is integral over R_1 and it follows that the ring $R_1[t] = R[a_0, a_1, \dots, a_{n-1}, t]$ is a finitely generated R -module. By the proposition, this means that t is integral over R , which gives (3).

The second statement in Corollary 24 shows that taking the elements of S that are integral over R gives a (possibly larger) subring of S , and the last statement in the corollary shows that the process of taking the integral closure stops after one step:

Corollary 25. Let R be a subring of the commutative ring S with $1 \in R$. Then the integral closure of R in S is integrally closed in S .

Examples

- (1) If R and S are fields then S is integral over R if and only if S is algebraic over R — if $s \in S$ is a root of the polynomial $p(x)$ with coefficients in R then it is a root of the monic polynomial obtained by dividing by the (nonzero) leading coefficient of $p(x)$.
- (2) Suppose S is an integral extension of R and I is an ideal in S . Then S/I is an integral ring extension of $R/(R \cap I)$ (reducing the monic polynomial over R satisfied by $s \in S$ modulo I gives a monic polynomial satisfied by $\bar{s} \in S/I$ over $R/(R \cap I)$).
- (3) If R is a U.F.D. then R is integrally closed, as follows. Suppose a/b is an element in the field of fractions of R (with $b \neq 0$ and a and b having no common factors) and satisfies $(a/b)^n + r_{n-1}(a/b)^{n-1} + \cdots + r_1(a/b) + r_0 = 0$ with $r_0, \dots, r_{n-1} \in R$. Then

$$a^n = b(-r_{n-1}a^{n-1} - \cdots - r_1ab^{n-2} - r_0b^{n-1})$$

shows that any irreducible element dividing b divides a^n , hence divides a . Since a/b is in lowest terms, this shows that b must be a unit, i.e., $a/b \in R$.

- (4) The polynomial ring $k[x, y]$ over the field k is integrally closed in its fraction field $k(x, y)$ by example (3) above. The ideal $(x^2 - y^3)$ is prime (cf. Exercise 14, Section 9.1), so the quotient ring $R = k[x, y]/(x^2 - y^3) = k[\bar{x}, \bar{y}]$ is an integral domain. This domain is not integrally closed, however, since $\bar{x}/\bar{y}$ is an element of the fraction field of R that is integral over R (since $(\bar{x}/\bar{y})^3 - \bar{x} = 0$), but is not an element of R . In particular, R is not a U.F.D. by the previous example.

We next consider the behavior of ideals in integral ring extensions.

Definition. Let $\varphi : R \rightarrow S$ be a homomorphism of commutative rings.

- (a) If I is an ideal in R then the *extension* of I to S is the ideal $\varphi(I)S$ of S generated by the image of I .
- (b) If J is an ideal of S , then the *contraction* in R of J is the ideal $\varphi^{-1}(J)$.

In the special case where R is a subring of S and φ is the natural injection, the extension of $I \subseteq R$ is the ideal IS in S and the contraction of $J \subseteq S$ is the ideal $J \cap R$ of R .

It is immediate from the definition that

- (1) $I \subseteq IS \cap R$, more generally, I is contained in the contraction of its extension to S , and
- (2) $(J \cap R)S \subseteq J$, more generally, J contains the extension of its contraction in R .

In general equality need not hold in either situation (cf. the exercises).

If Q is a prime ideal in S , then its contraction is prime in R (although the contraction of a maximal ideal need not be maximal). On the other hand, if P is a prime ideal in R , its extension need not be prime (or even proper) in S ; moreover, it is not generally true that P is the contraction of a prime ideal of S (cf. the exercises). For integral ring extensions, however, the situation is more controlled:

Theorem 26. Let R be a subring of the commutative ring S with $1 \in R$ and suppose that S is integral over R .

- (1) Assume that S is an integral domain. Then R is a field if and only if S is a field.
- (2) Let P be a prime ideal in R . Then there is a prime ideal Q in S with $P = Q \cap R$. Moreover, P is maximal if and only if Q is maximal.
- (3) (*The Going-up Theorem*) Let $P_1 \subseteq P_2 \subseteq \cdots \subseteq P_n$ be a chain of prime ideals in R and suppose there are prime ideals $Q_1 \subseteq Q_2 \subseteq \cdots \subseteq Q_m$ of S with $P_i = Q_i \cap R$, $1 \leq i \leq m$ and $m < n$. Then the ascending chain of ideals can be completed: there are prime ideals $Q_{m+1} \subseteq \cdots \subseteq Q_n$ in S such that $P_i = Q_i \cap R$ for all i .
- (4) (*The Going-down Theorem*) Assume that S is an integral domain and R is integrally closed in S . Let $P_1 \supseteq P_2 \supseteq \cdots \supseteq P_n$ be a chain of prime ideals in R and suppose there are prime ideals $Q_1 \supseteq Q_2 \supseteq \cdots \supseteq Q_m$ of S with $P_i = Q_i \cap R$, $1 \leq i \leq m$ and $m < n$. Then the descending chain of ideals can be completed: there are prime ideals $Q_{m+1} \supseteq \cdots \supseteq Q_n$ in S such that $P_i = Q_i \cap R$ for all i .

Proof: To prove (1) assume first that R is a field and let s be a nonzero element of S . Then s is integral over R , so

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0 = 0$$

for some $a_0, a_1, \dots, a_{n-1}$ in R . Since S is an integral domain, we may assume $a_0 \neq 0$ (otherwise cancel factors of s). Then

$$s(s^{n-1} + a_{n-1}s^{n-2} + \cdots + a_1) = -a_0$$

and since $(-1/a_0) \in R$, this shows that $(-1/a_0)(s^{n-1} + a_{n-1}s^{n-2} + \cdots + a_1)$ is an inverse for s in S , so S is a field. Conversely, suppose S is a field and r is a nonzero element of R . Since $r^{-1} \in S$ is integral over R we have

$$r^{-m} + a_{m-1}r^{-m+1} + \cdots + a_1r^{-1} + a_0 = 0$$

for some $a_0, \dots, a_{m-1} \in R$. Then $r^{-1} = -(a_{m-1} + \dots + a_1 r^{m-2} + a_0 r^{m-1}) \in R$, so R is a field.

The proof of the first statement in (2) is given in Corollary 50. For the second statement, observe that the integral domain S/Q is an integral extension of R/P (Example 2 following Corollary 25). By (1), S/Q is a field if and only if R/P is a field, i.e., Q is maximal if and only if P is maximal.

To prove (3), it suffices by induction to prove that if $P_1 \subseteq P_2$ and Q_1 is a prime of S with $Q_1 \cap R = P_1$ then there is a prime Q_2 of S with $Q_1 \subseteq Q_2$ and $Q_2 \cap R = P_2$. Since $\bar{S} = S/Q_1$ is an integral extension of $\bar{R} = R/P_1$, the first part of (2) shows that there exists a prime $\bar{Q}_2$ of $\bar{S}$ with $\bar{Q}_2 \cap \bar{R} = P_2/P_1$. Then the preimage Q_2 of $\bar{Q}_2$ in S is a prime ideal containing Q_1 with $Q_2 \cap R = P_2$.

The proof of (4) is outlined in Exercise 24 in Section 4.

Corollary 27. Suppose R is a subring of the ring S with $1 \in R$ and assume S is integral and finitely generated (as a ring) over R . If P is a maximal ideal in R then there is a nonzero and finite number of maximal ideals Q of S with $Q \cap R = P$.

Proof: There exists at least one maximal ideal Q lying over P by (2) of the theorem, so we must see why there are only finitely many such maximal ideals in S . If Q is a maximal ideal of S with $Q \cap R = P$ then S/Q is a field containing the field R/P . To prove that there are only finitely many possible Q it suffices to prove that there are only finitely many homomorphisms from S to a field containing R/P that extend the homomorphism from R to R/P . Let $S = R[s_1, \dots, s_n]$, where the elements s_i are integral over R by assumption, and let $p_i(x)$ be a monic polynomial with coefficients in R satisfied by s_i . If Q is a maximal ideal of S then $S/Q = (R/P)[\bar{s}_1, \dots, \bar{s}_n]$ is the field extension of the field R/P with generators $\bar{s}_1, \dots, \bar{s}_n$. The element $\bar{s}_i$ is a root of the monic polynomial $\bar{p}_i(x)$ with coefficients in R/P obtained by reducing the coefficients of $p_i(x)$ mod P . There are only a finite number of possible roots of this monic polynomial (in a fixed algebraic closure of R/P), and so only finitely many possible field extensions of the form $(R/P)[\bar{s}_1, \dots, \bar{s}_n]$, which proves the corollary.

Algebraic Integers

We can use the concept of an integral ring extension to define the “integers” in extension fields of the rational numbers $\mathbb{Q}$:

Definition. Let K be an extension field of $\mathbb{Q}$.

- (1) An element $\alpha \in K$ is called an *algebraic integer* if α is integral over $\mathbb{Z}$, i.e., if α is the root of some monic polynomial with coefficients in $\mathbb{Z}$.
- (2) The integral closure of $\mathbb{Z}$ in K is called the *ring of integers* of K , and is denoted by $\mathcal{O}_K$.

An algebraic integer is clearly algebraic over $\mathbb{Q}$, so the ring of all algebraic integers is the ring of integers in $\bar{\mathbb{Q}}$, an algebraic closure of $\mathbb{Q}$. Examples of algebraic integers include $\sqrt{2}$, $\sqrt{-1}$, $\sqrt[3]{5}$, etc. since these elements are certainly roots of monic polynomials with coefficients in $\mathbb{Z}$. The definition of an algebraic integer α is that α be a root

of *some* monic polynomial in $\mathbb{Z}[x]$, a condition which seems difficult to check. The next proposition gives a simple criterion for α to be an algebraic integer in terms of the minimal polynomial for α .

Proposition 28. An element α in some field extension of $\mathbb{Q}$ is an algebraic integer if and only if α is algebraic over $\mathbb{Q}$ and its minimal polynomial $m_{\alpha, \mathbb{Q}}(x)$ has integer coefficients. In particular, the algebraic integers in $\mathbb{Q}$ are the integers $\mathbb{Z}$, i.e., $\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$.

Proof: If α is algebraic over $\mathbb{Q}$ with $m_{\alpha, \mathbb{Q}}(x) \in \mathbb{Z}[x]$, then by definition α is integral over $\mathbb{Z}$. Conversely, assume α is integral over $\mathbb{Z}$, and let $f(x)$ be a monic polynomial in $\mathbb{Z}[x]$ of minimum degree having α as a root. If f were reducible in $\mathbb{Q}[x]$, then by Gauss' Lemma $f(x) = g(x)h(x)$ for some monic polynomials $g(x), h(x)$ in $\mathbb{Z}[x]$ of degree smaller than the degree of f . But then α would be a root of either g or h , contradicting the minimality of f . Hence f is irreducible in $\mathbb{Q}[x]$, so $f(x) = m_{\alpha, \mathbb{Q}}(x)$ and so the minimal polynomial for α has coefficients in $\mathbb{Z}$. Finally, the minimal polynomial of $\alpha = a/b \in \mathbb{Q}$ (a/b reduced to lowest terms and $b > 0$) is $bx - a$, which is monic if and only if $b = 1$, so $\alpha \in \mathbb{Q}$ is an algebraic integer if and only if $\alpha \in \mathbb{Z}$.

Because the integers $\mathbb{Z}$ are the algebraic integers in $\mathbb{Q}$, for emphasis (and clarity) the elements of $\mathbb{Z}$ are sometimes referred to as the “rational integers” to distinguish them from the “integers” in extensions of finite degree over $\mathbb{Q}$ (called *number fields*). The next result gives some of the basic structure of the ring of integers in a general number field.

Theorem 29. Let K be a number field of degree n over $\mathbb{Q}$.

- (1) The ring $\mathcal{O}_K$ of integers in K is a Noetherian ring and is a free $\mathbb{Z}$ -module of rank n .
- (2) For every $\beta \in K$ there is some nonzero $d \in \mathbb{Z}$ such that $d\beta$ is an algebraic integer. In particular, K is the field of fractions of $\mathcal{O}_K$.
- (3) If $\beta_1, \beta_2, \dots, \beta_n$ is any $\mathbb{Q}$ -basis of K , then there is an integer d such that $d\beta_1, d\beta_2, \dots, d\beta_n$ is a basis for a free $\mathbb{Z}$ -submodule of $\mathcal{O}_K$ of rank n . Any basis of the $\mathbb{Z}$ -module $\mathcal{O}_K$ is also a basis for K as a vector space over $\mathbb{Q}$.

Proof: Note first that any $\mathbb{Z}$ -linear dependence relation among elements in $\mathcal{O}_K$ is a $\mathbb{Q}$ -linear dependence relation in K , and multiplying a $\mathbb{Q}$ -linear dependence relation of elements of $\mathcal{O}_K$ in K by a common denominator for the coefficients yields a $\mathbb{Z}$ -linear dependence relation in $\mathcal{O}_K$. Let β be any element of K and let $x^k + a_{k-1}x^{k-1} + \dots + a_0$ be the minimal polynomial of β over $\mathbb{Q}$. If d is a common denominator for the coefficients, then multiplying through by d^k shows that

$$(d\beta)^k + da_{k-1}(d\beta)^{k-1} + \dots + d^{k-1}a_1(d\beta) + d^ka_0 = 0,$$

and $d^ka_0, d^{k-1}a_1, \dots, da_{k-1} \in \mathbb{Z}$. Hence $d\beta$ is an algebraic integer, which proves the first part of (2) and then the second statement in (2) follows immediately.

If $\beta_1, \dots, \beta_n$ are a $\mathbb{Q}$ -basis for K over $\mathbb{Q}$, then there is a nonzero integer d such that $d\beta_1, \dots, d\beta_n$ all lie in $\mathcal{O}_K$. These elements are still linearly independent over $\mathbb{Q}$, so in particular are independent over $\mathbb{Z}$, hence generate a free submodule of $\mathcal{O}_K$ of rank n ,

which proves the first statement in (3).

Since $\mathcal{O}_K$ is a subring of the field K , it is a torsion free $\mathbb{Z}$ -module. If $\mathcal{O}_K$ were contained in some finitely generated $\mathbb{Z}$ -module it would follow that $\mathcal{O}_K$ is also finitely generated over $\mathbb{Z}$, hence is a free $\mathbb{Z}$ -module. If L is the Galois closure of K , then $\mathcal{O}_K \subseteq \mathcal{O}_L$ and so it suffices to see that $\mathcal{O}_L$ is contained in a finitely generated $\mathbb{Z}$ -module. Let $\alpha_1, \dots, \alpha_m$ be a $\mathbb{Q}$ -basis for L . Multiplying by an integer $d \in \mathbb{Z}$, if necessary, we may assume that each α_i is an algebraic integer, i.e., $\alpha_1, \dots, \alpha_m \in \mathcal{O}_L$. For each fixed $\theta \neq 0$ in L , the map

$$T_\theta : L \rightarrow \mathbb{Q} \quad \text{defined by} \quad T_\theta(\alpha) = \text{Tr}_{L/\mathbb{Q}}(\theta\alpha)$$

(where $\text{Tr}_{L/\mathbb{Q}}$ denotes the trace map from L to $\mathbb{Q}$, cf. Exercise 18 in Section 14.2) is a $\mathbb{Q}$ -linear transformation from L to $\mathbb{Q}$. This linear transformation is nonzero because $T_\theta(\theta^{-1}) = \text{Tr}_{L/\mathbb{Q}}(1) = m$. It follows that the map from L to $\text{Hom}_{\mathbb{Q}}(L, \mathbb{Q})$ mapping θ to T_θ is an injective homomorphism of vector spaces over $\mathbb{Q}$. Since both spaces have the same dimension over $\mathbb{Q}$, the map is an isomorphism. Put another way, every linear functional on L is of the form T_θ for some $\theta \in L$. In particular, there are elements $\alpha'_1, \dots, \alpha'_m$ in L whose corresponding linear transformations $T_{\alpha'_i}$ give the dual basis of $\alpha_1, \dots, \alpha_m$, i.e.,

$$\text{Tr}_{L/\mathbb{Q}}(\alpha'_i \alpha_j) = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{otherwise.} \end{cases}$$

Since $\alpha'_1, \dots, \alpha'_m$ are linearly independent, they give a basis for L over $\mathbb{Q}$. Hence every element $\beta \in \mathcal{O}_L$ can be written

$$\beta = a_1 \alpha'_1 + \dots + a_i \alpha'_i + \dots + a_m \alpha'_m$$

with $a_1, \dots, a_m \in \mathbb{Q}$. Multiplying by α_j and taking the trace shows that

$$\text{Tr}_{L/\mathbb{Q}}(\beta \alpha_j) = a_1 \text{Tr}_{L/\mathbb{Q}}(\alpha'_1 \alpha_j) + \dots + a_i \text{Tr}_{L/\mathbb{Q}}(\alpha'_i \alpha_j) + \dots + a_m \text{Tr}_{L/\mathbb{Q}}(\alpha'_m \alpha_j) = a_j.$$

But β and α_j are both elements of $\mathcal{O}_L$, so also $\beta \alpha_j$ is an element of $\mathcal{O}_L$, and this implies that $a_j = \text{Tr}_{L/\mathbb{Q}}(\beta \alpha_j)$ is an element of $\mathbb{Z}$ (cf. Exercise 18(d) of Section 14.2). It follows that

$$\mathcal{O}_L \subseteq \mathbb{Z}\alpha'_1 + \dots + \mathbb{Z}\alpha'_m$$

so that $\mathcal{O}_L$ is contained in a finitely generated $\mathbb{Z}$ -module, proving that $\mathcal{O}_K$ (and also $\mathcal{O}_L$) is a free $\mathbb{Z}$ -module.

Since K has dimension n as a vector space over $\mathbb{Q}$, it follows that $\mathcal{O}_K$ is a free $\mathbb{Z}$ -module of rank at most n (by Theorem 5 of Section 12.1). Because $\mathcal{O}_K$ also contains a free $\mathbb{Z}$ -submodule of rank n , it follows that the $\mathbb{Z}$ -rank of $\mathcal{O}_K$ is precisely n , proving (1), and then the second statement in (3) follows by the remarks on $\mathbb{Z}$ -linear and $\mathbb{Q}$ -linear dependence relations.

Finally, any ideal I in $\mathcal{O}_K$ is a $\mathbb{Z}$ -submodule of a free $\mathbb{Z}$ -module of rank n , so is a free $\mathbb{Z}$ -module of rank at most n , and a set of $\mathbb{Z}$ -module generators for I is also a set of $\mathcal{O}_K$ -generators. Hence every ideal of $\mathcal{O}_K$ can be generated by at most n elements, which implies that $\mathcal{O}_K$ is a Noetherian ring and completes the proof.

Definition. An *integral basis* for the number field K is a basis of the ring of integers in K considered as a free $\mathbb{Z}$ -module of rank $[K : \mathbb{Q}]$.

If P is a nonzero prime ideal in the ring of integers $\mathcal{O}_K$ of a number field K then $P \cap \mathbb{Z}$ is a prime ideal in $\mathbb{Z}$. If $\alpha \in P$, then the constant term of the minimal polynomial for α over $\mathbb{Q}$ is then an element in $P \cap \mathbb{Z}$, which shows that $P \cap \mathbb{Z} = p\mathbb{Z}$ is also a nonzero prime ideal in $\mathbb{Z}$. By Theorem 26, every prime ideal (p) in $\mathbb{Z}$ arises in this way. Since $p\mathbb{Z}$ is a maximal ideal, it also follows from (2) in Theorem 26 that *nonzero prime ideals in $\mathcal{O}_K$ are maximal*, and then by Corollary 27, there are finitely many prime ideals P in $\mathcal{O}_K$ with $P \cap \mathbb{Z} = p\mathbb{Z}$. We shall see later (Corollary 16 in Section 16.3) that *every nonzero ideal in the ring of integers of a number field can be written uniquely as the product of prime ideals*, and in the case of the ideal $p\mathcal{O}_K$ the distinct prime factors are precisely the finitely many ideals P in $\mathcal{O}_K$ with $P \cap \mathbb{Z} = p\mathbb{Z}$. This property replaces the unique factorization of *elements* in $\mathcal{O}_K$ into primes (which need not hold since $\mathcal{O}_K$ need not be a U.F.D.). We shall also see that primary ideals in $\mathcal{O}_K$ are powers of prime ideals (in fact this is equivalent to the unique factorization of ideals of $\mathcal{O}_K$ into products of prime ideals, cf. the exercises).

Example: (The Ring of Integers in Quadratic Extensions of $\mathbb{Q}$)

If K is a quadratic extension of $\mathbb{Q}$ then $K = \mathbb{Q}(\sqrt{D})$ for some squarefree integer D . Then

$$\mathcal{O}_{\mathbb{Q}(\sqrt{D})} = \mathbb{Z}[\omega] = \mathbb{Z} \cdot 1 + \mathbb{Z} \cdot \omega,$$

with integral basis $1, \omega$, where

$$\omega = \begin{cases} \sqrt{D}, & \text{if } D \equiv 2, 3 \pmod{4} \\ \frac{1 + \sqrt{D}}{2}, & \text{if } D \equiv 1 \pmod{4}. \end{cases}$$

This is the quadratic integer ring introduced in Section 7.1. Since ω satisfies $\omega^2 - D = 0$ (respectively, $\omega^2 - \omega + (1 - D)/4 = 0$ for $D \equiv 2, 3 \pmod{4}$ (respectively, $D \equiv 1 \pmod{4}$), it follows that ω is an algebraic integer in K and so $\mathbb{Z}[\omega] \subseteq \mathcal{O}_K$. To prove that this is the full ring of integers in K , let $\alpha = a + b\sqrt{D}$ with $a, b \in \mathbb{Q}$, and suppose that α is an algebraic integer. If $b = 0$, then $\alpha \in \mathbb{Q}$ and so $a \in \mathbb{Z}$. If $b \neq 0$, the minimal polynomial of α is $x^2 - 2ax + (a^2 - b^2D)$. Then Proposition 28 shows that $2a$ and $a^2 - b^2D$ are elements of $\mathbb{Z}$. Then $4(a^2 - b^2D) = (2a)^2 - (2b)^2D \in \mathbb{Z}$, hence $4b^2D \in \mathbb{Z}$. Since D is squarefree it follows that $2b$ is an integer. Write $a = x/2$ and $b = y/2$ for some integers x, y . Since $a^2 - b^2D$ is an integer, $x^2 - y^2D \equiv 0 \pmod{4}$. Since 0 and 1 are the only squares mod 4 and D is not divisible by 4, it is easy to check that the only possibilities are the following:

- (i) $D \equiv 2$ or $3 \pmod{4}$ and x, y are both even, or
- (ii) $D \equiv 1 \pmod{4}$ and x, y are both even or both odd.

In case (i), $a, b \in \mathbb{Z}$ and $\alpha \in \mathbb{Z}[\omega]$. In case (ii), $a + b\sqrt{D} = r + s\omega$ where $r = (x - y)/2$ and $s = y$ are both integers, so again $\alpha \in \mathbb{Z}[\omega]$.

Example: (The Ring of Integers in Cyclotomic Fields)

The ring of integers in the cyclotomic field $\mathbb{Q}(\zeta_n)$ of n^{th} roots of unity is $\mathbb{Z}[\zeta_n]$, where ζ_n is any primitive n^{th} root of 1. The elements $1, \zeta_n, \dots, \zeta_n^{\varphi(n)-1}$ are an integral basis. It is clear that ζ_n is an algebraic integer since it is a root of $x^n - 1$, so the ring $\mathbb{Z}[\zeta_n]$ is contained in the ring of integers. The proof that this is the full ring of algebraic integers in $\mathbb{Q}(\zeta_n)$ involves techniques from algebraic number theory beyond the scope of the material here.

Noether's Normalization Lemma and Hilbert's Nullstellensatz

We now apply some of the techniques from the algebraic theory of integral ring extensions to affine geometry.

Definition. If k is a field the elements $y_1, y_2, \dots, y_q$ in some k -algebra are called *algebraically independent* over k if there is no nonzero polynomial p in q variables over k such that $p(y_1, y_2, \dots, y_q) = 0$.

Thus $y_1, y_2, \dots, y_q$ are algebraically independent if and only if the k -algebra homomorphism from the polynomial ring $k[x_1, \dots, x_q]$ to $k[y_1, \dots, y_q]$ defined by $x_i \mapsto y_i$ is an isomorphism. Elements in a field extension of k are algebraically independent if and only if they are independent transcendentals over k .

Theorem 30. (Noether's Normalization Lemma) Let k be a field and suppose that $A = k[r_1, r_2, \dots, r_m]$ is a finitely generated k -algebra. Then for some q , $0 \leq q \leq m$, there are algebraically independent elements $y_1, y_2, \dots, y_q \in A$ such that A is integral over $k[y_1, y_2, \dots, y_q]$.

Proof: Proceed by induction on m . If $r_1, \dots, r_m$ are algebraically independent over k then take $y_i = r_i$, $i = 1, \dots, m$. Otherwise, there exists $f(x_1, \dots, x_m) \in k[x_1, \dots, x_m]$ such that $f(r_1, \dots, r_m) = 0$. The polynomial f is a sum of monomials of the form $ax_1^{e_1}x_2^{e_2}\cdots x_m^{e_m}$, where the degree of this monomial is $e_1 + \cdots + e_m$ and the degree, d , of f is the maximum of the degrees of its monomials. Renumbering the variables if necessary, we may assume that f is a nonconstant polynomial in x_m with coefficients in the ring $k[x_1, x_2, \dots, x_{m-1}]$. We now perform a change of variables that transforms (or "normalizes") f into a *monic* polynomial in x_m with coefficients from a subring of A which is generated over k by $m - 1$ elements, at which point we shall be able to apply induction.

Define integers $\alpha_i = (1 + d)^i$ and new variables $X_i = x_i - x_m^{\alpha_i}$ for $1 \leq i \leq m - 1$. Let

$$g(X_1, X_2, \dots, X_{m-1}, x_m) = f(X_1 + x_m^{\alpha_1}, X_2 + x_m^{\alpha_2}, \dots, X_{m-1} + x_m^{\alpha_{m-1}}, x_m),$$

so $g \in k[X_1, \dots, X_{m-1}, x_m]$. Each monomial term of f contributes a single term of the form a constant times x_m^e to g . It is also easy to check that the choice of α_i ensures that distinct monomials in f give different values of e (for example by viewing the degrees of the monomials in the new variables as integers expressed in base $b = d + 1$). If N is the highest power of x_m that occurs, then it follows that

$$g = cx_m^N + \sum_{i=0}^{N-1} h_i(X_1, \dots, X_{m-1})x_m^i$$

for some nonzero $c \in k$. If now $s_i = r_i - r_m^{\alpha_i}$ then

$$\frac{1}{c}g(s_1, s_2, \dots, s_{m-1}, r_m) = \frac{1}{c}f(r_1, r_2, \dots, r_{m-1}, r_m) = 0,$$

which shows that r_m is integral over $B = k[s_1, \dots, s_{m-1}]$. Each r_i for $1 \leq i \leq m - 1$ is integral over $B[r_m]$ since r_i is a root of the monic polynomial $x - s_i - r_m^{\alpha_i}$, so A is

integral over $B[r_m]$. By transitivity of integrality, A is therefore integral over B . Since B is a k -algebra generated by $m - 1$ elements, induction completes the proof.

A more “geometric” interpretation of Noether’s Normalization Lemma is indicated in Exercise 15. We next use the Normalization Lemma to prove that if k is an algebraically closed field then the maximal ideals of the polynomial ring $k[x_1, x_2, \dots, x_n]$ are of the form $(x_1 - a_1, \dots, x_n - a_n)$ for some $a_1, \dots, a_n \in k$. Viewing $k[x_1, x_2, \dots, x_n]$ as the ring of polynomial functions on $\mathbb{A}^n$, this says that the maximal ideals correspond to the kernels of evaluation maps at points of $\mathbb{A}^n$ — similar to the corresponding result for rings of continuous functions on a compact set (cf. Exercises 33, 34 in Section 7.4).

Theorem 31. (Hilbert’s Nullstellensatz — Weak Form) Let k be an algebraically closed field. Then M is a maximal ideal in the polynomial ring $k[x_1, x_2, \dots, x_n]$ if and only if $M = (x_1 - a_1, \dots, x_n - a_n)$ for some $a_1, \dots, a_n \in k$. Equivalently, the maps $\mathcal{Z}$ and $\mathcal{I}$ give a bijective correspondence

$$\{\text{points in } \mathbb{A}^n\} \begin{array}{c} \xrightarrow{\mathcal{I}} \\ \xleftarrow{\mathcal{Z}} \end{array} \{\text{maximal ideals in } k[\mathbb{A}^n]\}.$$

Moreover, if I is any proper ideal in $k[x_1, x_2, \dots, x_n]$ then $\mathcal{Z}(I) \neq \emptyset$.

Proof: Certainly $(x_1 - a_1, \dots, x_n - a_n)$ is a maximal ideal in $k[x_1, x_2, \dots, x_n]$. Conversely, for any maximal ideal M in $k[x_1, x_2, \dots, x_n]$, let $E = k[x_1, x_2, \dots, x_n]/M$. Then E is a field containing k that is finitely generated over k (by $\bar{x}_1, \dots, \bar{x}_n$). By Noether’s Normalization Lemma, E is integral over a polynomial ring $k[y_1, \dots, y_q]$. Then $k[y_1, \dots, y_q]$ is a field by Theorem 26(1), and since a polynomial ring in one or more variables is never a field, it follows that $q = 0$. Hence E is integral over k , so E is algebraic over k . Because k is algebraically closed, $E = k$, i.e., $\bar{x}_i \in k$ for $1 \leq i \leq n$. Hence for $i = 1, \dots, n$ there is some $a_i \in k$ such that $x_i - a_i \in M$. This means that the maximal ideal $(x_1 - a_1, \dots, x_n - a_n)$ is contained in M , so $M = (x_1 - a_1, \dots, x_n - a_n)$. Finally, if I is any nonzero ideal in $k[x_1, x_2, \dots, x_n]$ then I is contained in a maximal ideal $M = (x_1 - a_1, \dots, x_n - a_n)$, and so $(a_1, \dots, a_n) \in \mathcal{Z}(I)$.

Theorem 32. (Hilbert’s Nullstellensatz) Let k be an algebraically closed field. Then $\mathcal{I}(\mathcal{Z}(I)) = \text{rad } I$ for every ideal I of $k[x_1, x_2, \dots, x_n]$. Moreover, the maps $\mathcal{Z}$ and $\mathcal{I}$ define inverse bijections

$$\{\text{affine algebraic sets}\} \begin{array}{c} \xrightarrow{\mathcal{I}} \\ \xleftarrow{\mathcal{Z}} \end{array} \{\text{radical ideals}\}.$$

Proof: Since $\text{rad } I \subseteq \mathcal{I}(\mathcal{Z}(I))$ it remains to prove the reverse inclusion. By Hilbert’s Basis Theorem, $I = (f_1, f_2, \dots, f_m)$. Let $g \in \mathcal{I}(\mathcal{Z}(I))$. Introduce a new variable x_{n+1} and consider the ideal I' generated by $f_1, \dots, f_m$ and $x_{n+1}g - 1$ in $k[x_1, \dots, x_n, x_{n+1}]$. At any point of $\mathbb{A}^{n+1}$ where $f_1, \dots, f_m$ vanish the polynomial g also vanishes since $g \in \mathcal{I}(\mathcal{Z}(I))$, so that $x_{n+1}g - 1$ is nonzero. Hence $\mathcal{Z}(I') = \emptyset$ in $\mathbb{A}^{n+1}$. By the Weak Form of the Nullstellensatz, I' cannot be a proper ideal, i.e., $1 \in I'$. Write

$$1 = a_1 f_1 + \dots + a_m f_m + a_{m+1}(x_{n+1}g - 1) \quad \text{for some } a_i \in k[x_1, \dots, x_{n+1}].$$

Letting $y = 1/x_{n+1}$ and multiplying by a high power of y in this equation shows that

$$y^N = c_1 f_1 + \cdots + c_m f_m + c_{m+1}(g - y) \quad \text{for some } c_i \in k[x_1, \dots, x_n, y].$$

Substituting g for y in this polynomial equation shows that $g^N \in I$ (in $k[x_1, \dots, x_n]$), i.e., $g \in \text{rad } I$. Hence $\mathcal{I}(\mathcal{Z}(I)) \subseteq \text{rad } I$ and so $\mathcal{I}(\mathcal{Z}(I)) = \text{rad } I$, completing the proof.

It follows directly from Proposition 12 and Theorem 26(2) that if S is an integral extension of R with $1 \in R$ and if I is an ideal of R , then

$$(\text{rad}_S IS) \cap R = \text{rad}_R I$$

where IS is the ideal generated by I in S , and the subscript indicates the ring in which the radicals are being computed. This has the following geometric interpretation.

Corollary 33. (*Variant of Hilbert's Nullstellensatz*) If k is any field with algebraic closure $\bar{k}$ and I is an ideal in $k[x_1, x_2, \dots, x_n]$, then $\mathcal{I}_k(\mathcal{Z}_{\bar{k}}(I)) = \text{rad } I$, where $\mathcal{Z}_{\bar{k}}(I)$ is the zero set in $\bar{k}^n$ of the polynomials in I and $\mathcal{I}_k(\mathcal{Z}_{\bar{k}}(I))$ is the ideal of polynomials in $k[x_1, x_2, \dots, x_n]$ vanishing at all the points in $\mathcal{Z}_{\bar{k}}(I)$. In particular, $I = (1)$ if and only if there are no common zeros in $\bar{k}^n$ of the polynomials in I .

Proof: Since $\bar{k}[x_1, x_2, \dots, x_n]$ is an integral extension of $k[x_1, x_2, \dots, x_n]$ (generated by the integral elements $\bar{k}$), the corollary follows immediately from Theorem 32 and the remarks on radicals above.

From the Nullstellensatz we now have a dictionary between geometric and ring-theoretic objects over the algebraically closed field k :

Geometry	Algebra
affine algebraic set V	coordinate ring $k[V]$
points of V	maximal ideals of $k[V]$
affine algebraic subsets in V	radical ideals of $k[V]$
subvarieties in V	prime ideals in $k[V]$
morphism $\varphi : V \rightarrow W$	k -algebra homomorphism $\tilde{\varphi} : k[W] \rightarrow k[V]$

Computing Radicals

There are algorithms for computing radicals and primary decompositions in polynomial rings using Gröbner bases. While they are relatively elementary, they are somewhat technical and so we limit our discussion here to some preliminary results.

For hypersurfaces $V = \mathcal{Z}(f)$ defined by a single polynomial $f \in k[x_1, \dots, x_n]$, determining $\mathcal{I}(V) = \text{rad}(f)$ is straightforward. Since $k[x_1, \dots, x_n]$ is a U.F.D., f factors uniquely as the product of powers of nonassociate irreducibles: $f = p_1^{a_1} \cdots p_s^{a_s}$ and then $\text{rad}(f)$ is generated by $p_1 \cdots p_s$ (the ‘squarefree part’ of f).

Example

Suppose $W = \mathcal{Z}(J)$ with $J = (u^3 - uv^2 + v^3) \in \mathbb{Q}[u, v]$. The polynomial $x^3 - x + 1$ is irreducible over $\mathbb{Q}$, so $f = u^3 - uv^2 + v^3$ is irreducible in $\mathbb{Q}[u, v]$. Hence $\text{rad } J = J$ and $\mathcal{I}(W) = J$.

For nonprincipal ideals I , determining $\text{rad } I$ is more complicated. The following proposition (based on Hilbert's Nullstellensatz) gives a criterion determining when an element is contained in $\text{rad } I$.

Proposition 34. Suppose k is any field. If $I = (f_1, \dots, f_s)$ is a proper ideal in $k[x_1, \dots, x_n]$, then $f \in \text{rad } I$ if and only if $(f_1, \dots, f_s, 1 - yf) = k[x_1, \dots, x_n, y]$.

Proof: By Corollary 33, $(f_1, \dots, f_s, 1 - yf) = k[x_1, \dots, x_n, y]$ if and only if the equations

$$1 - yf(x_1, \dots, x_n) = 0, \quad f_1(x_1, \dots, x_n) = 0, \quad \dots, \quad f_s(x_1, \dots, x_n) = 0$$

have no common zero over the algebraic closure $\bar{k}$ of k . For a given $(a_1, \dots, a_n) \in \bar{k}^n$, the equation $1 - yf(a_1, \dots, a_n) = 0$ has a solution y unless $f(a_1, \dots, a_n) = 0$. Hence, the system of equations has no common zero if and only if for every $(a_1, \dots, a_n) \in \bar{k}^n$ with $f_1(a_1, \dots, a_n) = \dots = f_s(a_1, \dots, a_n) = 0$ we also have $f(a_1, \dots, a_n) = 0$. Equivalently, if $(a_1, \dots, a_n) \in \mathcal{Z}_{\bar{k}}(I)$, then also $f(a_1, \dots, a_n) = 0$, i.e., we have $f \in \mathcal{I}_k(\mathcal{Z}_{\bar{k}}(I)) = \text{rad } I$, by Corollary 33.

Since the reduced Gröbner basis (with respect to any fixed monomial ordering) for an ideal is unique, we immediately obtain the following algorithmic method for determining when a polynomial lies in the radical of an ideal.

Corollary 35. Suppose $I = (f_1, \dots, f_s)$ in $k[x_1, \dots, x_n]$. Then $f \in \text{rad } I$ if and only if $\{1\}$ is the reduced Gröbner basis for the ideal $(f_1, \dots, f_s, 1 - yf)$ in $k[x_1, \dots, x_n, y]$ with respect to any monomial ordering.

Example

Consider $I = (x^2 - y^2, xy)$ in $k[x, y]$. The reduced Gröbner basis for $(x^2 - y^2, xy, 1 - tx)$ in $k[x, y, t]$ with respect to the order $x > y > t$ is $\{1\}$, showing $x \in \text{rad}(I)$. To determine the smallest power of x lying in I , we find that the ideal $(x^2 - y^2, xy, x^3)$ in $k[x, y]$ has the same reduced Gröbner basis as I (namely $\{x^2 - y^2, xy, y^3\}$), but $(x^2 - y^2, x^2, xy)$ has basis $\{x^2, xy, y^2\}$. It follows that $x^3 \in I$ and $x^2 \notin I$ (alternatively, x^3 leaves a nonzero remainder after general polynomial division by $\{x^2 - y^2, xy, y^3\}$, but x^3 has a remainder of 0). By a similar computation (or by symmetry), $y \in \text{rad } I$, with $y^3 \in I$ but $y^2 \notin I$. Since $(x, y) \subseteq \text{rad } I$, it follows that $\text{rad } I = (x, y)$.

Some additional results for computing radicals are presented in the exercises.

EXERCISES

Let R be a subring of the commutative ring S with $1 \in R$.

1. Use the fact that a U.F.D. is integrally closed to prove that the Gaussian integers, $\mathbb{Z}[i]$, is the ring of integers in $\mathbb{Q}(i)$.
2. Suppose k is a field and let $t = \bar{x}/\bar{y}$ in the field of fractions of the integral domain $R = k[x, y]/(x^2 - y^3)$. Prove that $K = k(t)$ is the fraction field of R and $k[t]$ is the integral closure of R in K .
3. Suppose k is a field and i and j are relatively prime positive integers. Find the normalization of the integral domain $R = k[x, y]/(x^i - y^j)$ (cf. Exercise 14, Section 9.1).
4. Suppose k is a field and let P be the ideal $(y^2 - x^3 - x^2)$ in the polynomial ring $k[x, y]$. Prove that P is a prime ideal and find the normalization of the integral domain $R = k[x, y]/P$. [To prove P is prime, show that $y^2 - x^3 - x^2$ is irreducible in the U.F.D. $k[x, y]$. Then consider $t = \bar{y}/\bar{x} \in R$.]
5. If R is an integral domain with field of fractions F , show that F is a finitely generated R -module if and only if $R = F$.
6. For each of the following give specific rings $R \subseteq S$ and explicit ideals in these rings that exhibit the specified relation:
 - (a) an ideal I of R such that $I \neq SI \cap R$ (so the contraction of the extension of an ideal I need not equal I)
 - (b) a prime ideal P of R such that there is no prime ideal Q of S with $P = Q \cap R$
 - (c) a maximal ideal M of S such that $M \cap R$ is not maximal in R
 - (d) a prime ideal P of R whose extension PS to S is not a prime ideal in S
 - (e) an ideal J of S such that $J \neq (J \cap R)S$ (so the extension of the contraction of an ideal J need not equal J).
7. Let $\mathcal{O}_K$ be the ring of integers in a number field K .
 - (a) Suppose that every nonzero ideal I of $\mathcal{O}_K$ can be written as the product of powers of prime ideals. Prove that an ideal Q of $\mathcal{O}_K$ is P -primary if and only if $Q = P^m$ for some $m \geq 1$. [Show first that since nonzero primes in $\mathcal{O}_K$ are maximal that $P_1^{m_1} \subseteq P_2^{m_2}$ for distinct nonzero primes P_1, P_2 implies $P_1 = P_2$.]
 - (b) Suppose that an ideal Q of $\mathcal{O}_K$ is P -primary if and only if $Q = P^m$ for some $m \geq 1$. Assuming all of Theorem 21, prove that every nonzero ideal I of $\mathcal{O}_K$ can be written uniquely as the product of powers of prime ideals. [Prove that $P_1^{m_1}$ and $P_2^{m_2}$ are comaximal ideals if P_1 and P_2 are distinct nonzero prime ideals and use the Chinese Remainder Theorem.]
8. Prove that if $s_1, \dots, s_n \in S$ are integral over R , then the ring $R[s_1, \dots, s_n]$ is a finitely generated R -module.
9. Suppose that S is integral over R and that P is a prime ideal in R . Prove that every element s in the ideal PS generated by P in S satisfies an equation $s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0 = 0$ where the coefficients $a_0, a_1, \dots, a_{n-1}$ are elements of P . [If $s = p_1s_1 + \dots + p_ms_m \in PS$, show that $T = R[s_1, \dots, s_m]$ satisfies the hypotheses in Proposition 23(3). Follow the proof in Proposition 23 that s is integral, noting that $s \in PT$ so that the a_{ij} are elements of P .]
10. Prove the following generalization of Proposition 28: Suppose R is an integrally closed integral domain with field of fractions k and α is an element of an extension field K of k . Show that α is integral over R if and only if α is algebraic over k and the minimal polynomial $m_{\alpha, k}(x)$ for α over k has coefficients in R . [If α is integral prove the conjugates

of α , i.e., the roots of $m_{\alpha,k}(x)$, are also integral, so the elementary symmetric functions of the conjugates are elements of k that are integral over R .]

11. Suppose R is an integrally closed integral domain with field of fractions k and $p(x) \in R[x]$ is a monic polynomial. Show that if $p(x) = a(x)b(x)$ with monic polynomials $a(x), b(x) \in k[x]$ then $a(x), b(x) \in R[x]$ (compare to Gauss' Lemma, Proposition 5, Section 9.3). [See the previous exercise.]
12. Suppose S is an integral domain that is integral over a ring R as in the previous exercise. If P is a prime ideal in R , let s be any element in the ideal PS generated by P in S . Prove that, with the exception of the leading term, the coefficients of the minimal polynomial $m_{s,k}(x)$ for s over k are elements of P . [By Exercise 10, $m_{s,k}(x) \in R[x]$. Exercise 9 shows that s is a root of a monic polynomial $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ with $a_0, \dots, a_{n-1} \in P$. Use the previous exercise to show that $p(x) = m_{s,k}(x)b(x)$ with $b(x)$ in $R[x]$, and consider this equation in the integral domain $(R/P)[x]$.]

The next two exercises extend Exercise 6 in Section 7.5 by characterizing fields that are not fields of fractions of any of their proper subrings.

13. Let K be a field of characteristic 0 and let A be a subring of K maximal with respect to $1/2 \notin A$. (Such A exists by Zorn's Lemma.) Let F be the field of fractions of A in K .
 - (a) Show that K is algebraic over F . [If t is transcendental over F , show that $1/2 \notin A[t]$.]
 - (b) Show that A is integrally closed in K . [Show that $1/2$ is not in the integral closure of A in K .]
 - (c) Deduce from (a) and (b) that $K = F$.
14. Show that a field K is the field of fractions of some proper subring of K if and only if K is not a subfield of the algebraic closure of a finite field. [If K contains t transcendental over $\mathbb{F}_p$ argue as in the preceding exercise with $1/t$ in place of $1/2$ to show that K is the quotient field of some proper subring.]

The next exercise gives a "geometric" interpretation of Noether's Normalization Lemma, showing that every affine algebraic set is a *finite covering* of some affine n -space.

15. Let V be an affine algebraic set over an algebraically closed field k . Prove that for some n there is a surjective morphism from V onto $\mathbb{A}^n$ with finite fibers, and that if V is a variety, then n can be taken to be the dimension of V . [By Noether's Normalization Lemma the finitely generated k -algebra $S = k[V]$ contains a polynomial subalgebra $R = k[x_1, x_2, \dots, x_n]$ such that S is integral over R . Apply Theorem 6 to the inclusion of R in S to obtain a morphism φ from V to $\mathbb{A}^n$. To see that φ is surjective with finite fibers, apply Corollary 27 to the maximal ideal $(x_1 - a_1, \dots, x_n - a_n)$ of R corresponding to a point $(a_1, \dots, a_n)$ of $\mathbb{A}^n$.]
16. Let V be an affine algebraic set in $\mathbb{C}^n$. Prove that V is compact in the Euclidean topology (i.e., closed and bounded) if and only if it is finite. [Use Exercise 18 in Section 2, the previous exercise, and the behavior of compact sets with respect to continuous functions.]
17. Let R be a subring of the commutative ring S with $1_S \in R$ and suppose that S is integral over R . This exercise proves that R and S have the same *Krull dimension*, cf. Section 16.1.
 - (a) If $P_1 \subset P_2 \subset \cdots \subset P_n$ is a chain of distinct prime ideals in R prove that there is a chain $Q_1 \subset Q_2 \subset \cdots \subset Q_n$ of distinct prime ideals in S with $Q_i \cap R = P_i$.
 - (b) Prove conversely that if $Q_1 \subset Q_2 \subset \cdots \subset Q_n$ is a chain of distinct prime ideals in S and $P_i = Q_i \cap R$ then $P_1 \subset P_2 \subset \cdots \subset P_n$ is a chain of distinct prime ideals in R . [To prove the P_i are distinct, pass to a quotient and reduce the problem to showing that if Q is a nonzero prime ideal in the integral domain S then $Q \cap R$ is a nonzero prime

ideal in R . In this case, if $s \in Q$ is nonzero, show that the constant coefficient of a polynomial of minimal degree in $R[x]$ satisfied by s is a nonzero element in $Q \cap R$.]

18. Let $V = \mathcal{Z}(I)$ and $W = \mathcal{Z}(J)$ where I is the ideal $(uv + v) \subset \mathbb{C}[u, v]$ and J is the ideal $(-2y - y^2 + 2z + z^2, 2x - yz - z^2) \subset \mathbb{C}[x, y, z]$.
 - (a) Show that I and J are prime ideals. Conclude that $I = \mathcal{I}(V)$ and $J = \mathcal{I}(W)$ and that V and W are varieties.
 - (b) Show that the map $\varphi: V \rightarrow W$ defined by $\varphi((a_1, a_2)) = (a_1^2 + a_2, a_1 + a_2, a_1 - a_2)$ is an isomorphism.
19. Let $I = (x^3 + y^3 + z^3, x^2 + y^2 + z^2, (x + y + z)^3) \subset k[x, y, z]$. Use Gröbner bases to show that $x, y, z \in \text{rad } I$ if $\text{ch}(k) \neq 2, 3$.
20. Let $I = (x^3 + y^3 + z^3, xy + xz + yz, xyz) \subset k[x, y, z]$. Use Gröbner bases to show that $x, y, z \in \text{rad } I$.
21. Let $I = (x^4 + y^4 + z^4, x + y + z) \subset k[x, y, z]$.
 - (a) Use Gröbner bases to show that $xy + xz + yz \in \text{rad } I$ if $\text{ch}(k) \neq 2$ and determine the smallest power of $xy + xz + yz$ contained in I . Show that none of x, y or z is contained in $\text{rad } I$.
 - (b) If $J = (x^4 + y^4 + z^4, x + y + z, xy + xz + yz)$ show that the reduced Gröbner basis of J relative to the lexicographic ordering $x > y > z$ is $\{x + y + z, y^2 + yz + z^2\}$. Deduce that $k[x, y, z]/J \cong k[y, z]/(y^2 + yz + z^2)$ and that J is radical if $\text{ch}(k) \neq 3$.
 - (c) If $\text{ch}(k) \neq 2, 3$, show that $\text{rad } I = J$.
 - (d) If $\text{ch}(k) = 3$, show that $\text{rad } I = (x - y, y - z)$.
 - (e) If $\text{ch}(k) = 2$, show that $I = (x + y + z)$ is a prime, hence radical, ideal.
22. Let $I = (x^2y + z^3, x + y^3 - z, 2y^4z - yz^2 - z^3) \subset k[x, y, z]$. Use Gröbner bases to show that $x, y, z \in \text{rad } I$ and conclude that $\text{rad } I = (x, y, z)$. Show that x^9, y^7, z^9 are the smallest powers of x, y, z , respectively, lying in I .
23. Let $V = \mathcal{Z}(x^3 - x^2z - y^2z)$ and $W = \mathcal{Z}(x^2 + y^2 - z^2)$ in $\mathbb{C}^3$. Show that $\mathcal{I}(V) = (x^3 - x^2z - y^2z)$ and $\mathcal{I}(W) = (x^2 + y^2 - z^2)$ in $\mathbb{C}[x, y, z]$.
24. Let $V = \mathcal{Z}(x^3 + y^3 + 7z^3) \subset \mathbb{C}^3$. Show that $\mathcal{I}(V) = (x^3 + y^3 + 7z^3)$ in $\mathbb{C}[x, y, z]$.
25. Let $I = (xz + y^2 + z^2, xy - xz + yz - 2z^2)$ and let $K = I + (x^2 - 3y^2 + yz) \subset \mathbb{C}[x, y, z]$.
 - (a) By Exercise 46 in Section 1, there is an injective $\mathbb{C}$ -algebra homomorphism from $\mathbb{C}[x, y, z]/K$ to $\mathbb{C}[u, v]/(u^3 - uv^2 + v^3)$. Use this together with the example preceding Proposition 34 to prove that K is a radical ideal and deduce that $\text{rad } I \subseteq K$.
 - (b) Show that $\text{rad } I \subseteq (y, z)$.
 - (c) Show that $K \cap (y, z) = I$ and deduce that I is radical, so that $\mathcal{I}(V) = I$ if $V = \mathcal{Z}(I)$.
 - (d) Show that $y(x^2 - 3y^2 + yz)$ and $z(x^2 - 3y^2 + yz)$ are elements of I but none of y, z , or $x^2 - 3y^2 + yz$ is contained in I .
26. Let I be an ideal in $k[x_1, \dots, x_n]$. Prove that the following are equivalent (an ideal satisfying any of these conditions is called a *zero-dimensional ideal* because of (d)):
 - (a) The quotient $k[x_1, \dots, x_n]/I$ has finite dimension as a vector space over k .
 - (b) $I \cap k[x_i] \neq 0$ for each $i = 1, 2, \dots, n$.
 - (c) If G is any reduced Gröbner basis for I then for each $i = 1, \dots, n$, there is a $g_i \in G$ with leading term $x_i^{n_i}$ for some $n_i \geq 1$.
 - (d) The set of common zeros $\mathcal{Z}_{\bar{k}}(I)$ of the polynomials in I in an algebraic closure $\bar{k}$ of k is finite.

[For (a) implies (b) use the injection $k[x_i]/(I \cap k[x_i]) \hookrightarrow k[x_1, \dots, x_n]/I$. For (b) implies (c) note some $LT(g_i)$ divides the leading term of a generator for $I \cap k[x_i]$. For (c) implies (a)

- use Exercise 37 in Section 9.6. Show (b) implies (d). For (d) implies (b) show the product $m_{a_1,k}(x_i) \dots m_{a_N,k}(x_i)$ of the minimal polynomials of the i^{th} coordinates $a_1, \dots, a_N$ of the points in $\mathcal{Z}_{\bar{k}}(I)$ is a nonzero polynomial in $\mathcal{I}(\mathcal{Z}_{\bar{k}}(I))$ and apply Corollary 33.]
27. Let I be a zero-dimensional ideal in $k[x_1, \dots, x_n]$ and let I' be the ideal generated by I in $\bar{k}[x_1, \dots, x_n]$ where $\bar{k}$ is the algebraic closure of k . Let $\mathcal{Z}(I)$ be the zero set of I in k^n and let $\mathcal{Z}_{\bar{k}}(I)$ be the zero set of I (equivalently, of I') in $\bar{k}^n$.
- (a) Prove that $|\mathcal{Z}_{\bar{k}}(I)| = \dim_{\bar{k}} \bar{k}[x_1, \dots, x_n] / \text{rad } I'$. [Show that $\text{rad } I'$ is the product of the maximal ideals corresponding to the points in $V_{\bar{k}}$ and use the Chinese Remainder Theorem.]
- (b) Show $|\mathcal{Z}(I)| \leq \dim_k k[x_1, \dots, x_n] / I$. [One approach: use Exercise 43 in Section 1 and observe that $\dim_{\bar{k}} \bar{k}[x_1, \dots, x_n] / \text{rad } I' \leq \dim_{\bar{k}} \bar{k}[x_1, \dots, x_n] / I'$.]
28. Suppose I is a zero-dimensional ideal in $k[x_1, \dots, x_n]$, and suppose $I \cap k[x_i]$ is generated by the nonzero polynomial h_i (cf. Exercise 26). Let r_i be the product of the irreducible factors of h_i (the ‘squarefree part’ of h_i).
- (a) Prove that $I + (r_1, \dots, r_n) \subseteq \text{rad } I$.
- (b) (*Radicals of zero-dimensional ideals for perfect fields*) If k is a perfect field, prove that $\text{rad } I = I + (r_1, \dots, r_n)$. [Use induction on n . Write $r_1 = p_1 \dots p_t$ with distinct irreducibles p_i in $k[x_1]$. If $J = I + (r_1, \dots, r_n)$ show that $J = J_1 \cap \dots \cap J_t$ where $J_t = J + (p_t)$. Show for each i that reduction modulo p_i induces an isomorphism $k[x_1, \dots, x_n] / J_i \cong K[x_2, \dots, x_n] / J'_i$ where K is the extension field $k[x] / (p_i)$ and $J'_i \subseteq K[x_2, \dots, x_n]$ is the reduction of the ideal J_i modulo (p_i) . Use Exercise 11 of Section 13.5 to show that the image of r_j in $J'_i \cap K[x_j]$ remains a nonzero squarefree polynomial for each $j = 2, \dots, n$ since k is perfect. Conclude by induction that J'_i is a radical ideal. Deduce that J_i is a radical ideal, and finally that J is a radical ideal.]
- (c) Find the radicals of $(x^7 + x + y^3, x^4 + y^3 + y)$, $(x^3 - xy^2 + x, x^2y + y^3)$, and $(x^4 + y^3, x^3 - xy + y^2)$ in $\mathbb{Q}[x, y]$ and of $(x^2 + y^2z, x^2y^2 + z^3, y^2 + z^2)$ in $\mathbb{Q}[x, y, z]$.
- (d) Let $k = \mathbb{F}_p(t)$. Show that $I = (x^p + t, y^p - t)$ is a zero-dimensional ideal in $k[x, y]$ such that both $I \cap k[x]$ and $I \cap k[y]$ contain nonzero squarefree polynomials, but that I is not a radical ideal (so the result in (b) need not hold if k is not perfect). [Show that $x + y \in \text{rad } I$ but $x + y \notin I$.]

15.4 LOCALIZATION

The idea of “localization at a prime” in a ring is an extremely powerful and pervasive tool in algebra for isolating the behavior of the ideals in a ring. It is an algebraic analogue of the familiar idea of localizing at a point when considering questions of, for example, the differentiability of a function $f(x)$ on the real line. In fact one of the important applications (and also one of the original motivations for the development) of this technique is to translate such “local” properties in the geometry of affine algebraic spaces to corresponding properties of their coordinate rings.

We first consider a very general construction of “rings of fractions.” Let D be a multiplicatively closed subset of R containing 1 (i.e., $1 \in D$ and $ab \in D$ if $a, b \in D$). The next result constructs a new ring $D^{-1}R$ which is the “smallest” ring in which the elements of D become units. This generalizes the construction of rings of fractions in Section 7.5 by allowing D to contain zero or zero divisors, and so in this case R need not embed as a subring of $D^{-1}R$.